

(2011-PP)2.1 Which of the following can be concluded from the Rutherford scattering experiment ?

- (1) The nucleus of an atom consists of protons and neutrons.
- (2) The nucleus of an atom is very small compared to the size of the atom.
- (3) Electromagnetic waves emitted from atoms of gases are of specific frequencies.

- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

A	B	C	D
☐	☐	☐	☐

(2011-PP)2.2

The ionization energy for a hydrogen atom in ground state is 13.6 eV. A photon of energy 4.53 eV strikes a hydrogen atom in ground state. The hydrogen atom will

- A. not be excited to a higher energy level.
- B. be excited to the first excited state.
- C. be excited to the third excited state.
- D. be ionized.

A

B

C

D



(2011-PP)2.3

The wavelength of the radiation emitted when an electron of an atom drops from the j^{th} excited state of energy E_j to a lower k^{th} excited state of energy E_k is

A. $\frac{E_j - E_k}{h}$

B. $\frac{E_j - E_k}{hc}$

C. $\frac{hc}{E_j - E_k}$

D. $\frac{hc}{E_j} - \frac{hc}{E_k}$

A



B



C



D



(2011-PP)2.4 The de Broglie wavelength of a particle at speed v is λ . If the speed of the particle is doubled, the de Broglie wavelength is

A. $\lambda / 4$

B. $\lambda / 2$

C. λ

D. 2λ

A



B



C

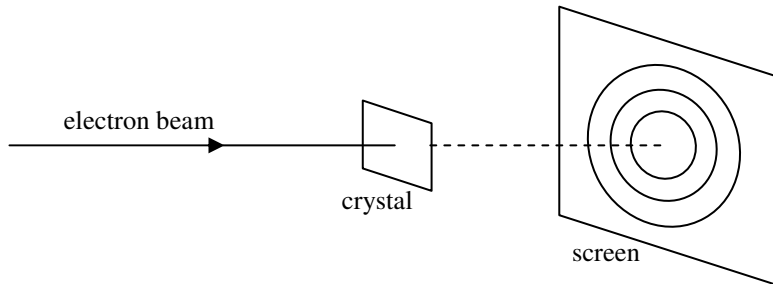


D



(2011-PP)2.5

A beam of electrons is incident on a thin film of crystal. A pattern of bright and dark rings is observed on a fluorescent screen. Which physical phenomenon explains the formation of the pattern ?



- A. Photoelectric effect
- B. Electron diffraction
- C. Ionization of atoms
- D. Lotus effect

A	B	C	D
☐	☐	☐	☐

(2011-PP)2.6

Which of the following statements about different microscopes is/are correct ?

- (1) The resolution of an optical microscope will increase if red light instead of blue light is used to illuminate the specimen.
- (2) A transmission electron microscope (TEM) uses magnetic field to focus the electron beam.
- (3) Only specimens that conduct electricity can be studied by a scanning tunnelling microscope (STM).

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2011-PP)2.7 Estimate the wavelength of electrons when they are accelerated in a transmission electron microscope (TEM) with a voltage of 76 kV.

A. $2.4 \times 10^{-12} \text{ m}$

B. $4.5 \times 10^{-12} \text{ m}$

C. $1.4 \times 10^{-10} \text{ m}$

D. $9.6 \times 10^{-9} \text{ m}$

A



B



C



D



(2011-PP)2.8

Which of the following are possible means by which nano particles could get into the human body?

- (1) The skin having direct contact with nano particles.
- (2) Inhaling nano particles into the lungs while breathing.
- (3) Ingesting food containing nano particles.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

A

B

C

D



(2011-PP)3.1

A 100 W filament light bulb and a 22 W compact fluorescent lamp both produce a luminous flux of 1600 lm. Which of the following statements about the two light sources is/are correct ?

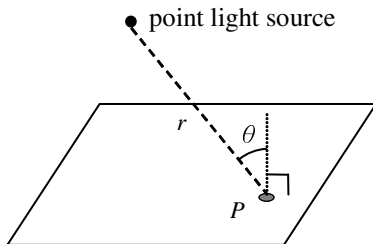
- (1) Both light sources give out the same amount of energy in the form of electromagnetic waves in 1 s.
- (2) 78 J of electrical energy is converted to heat in the filament light bulb in 1 s.
- (3) Both light sources have the same brightness to the human eye when observed from the same distance.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2011-PP)
3.2

A point light source with luminous flux F is illuminating a point P on a table surface as shown in the figure below. The illuminance at point P is



A. $\frac{F \cos \theta}{4\pi r^2}$

B. $\frac{F \cos^2 \theta}{4\pi r^2}$

C. $\frac{F \cos (90^\circ - \theta)}{4\pi r^2}$

D. $\frac{F \cos^2 (90^\circ - \theta)}{4\pi r^2}$

A



B



C



D



(2011-PP)
3.3

Which of the following statements about an electric hotplate and an induction cooker is/are correct ?

- (1) Both cookers make use of the heating effect of a current.
- (2) Only metal cooking pots can be used for both cookers.
- (3) In general, an induction cooker has a higher energy efficiency than an electric hotplate.

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

A

B

C

D



(2011-PP)3.4

The table below shows the data of a house. Calculate the Overall Thermal Transfer Value of the house.

	Windows	Walls	Roof
Rate of heat transfer / W	6200	4400	8600
Total area / m²	20	480	140

- A. 30 W m⁻²
- B. 127 W m⁻²
- C. 310 W m⁻²
- D. 381 W m⁻²

A

B

C

D



(2011-PP)3.5

A wind turbine has an overall efficiency of 30% and its output power is 360 kW when the wind blows normally at the turbine with a constant velocity of 10 m s^{-1} . Find the length of the blades of the wind turbine.

Given: density of air = 1.2 kg m^{-3}

- A. 4.1 m
- B. 7.6 m
- C. 13.8 m
- D. 25.2 m

A	B	C	D
☐	☐	☐	☐

(2011-PP)3.6

The difference in mass between a uranium-238 nucleus and its constituent nucleons is 1.88482 u.
Determine the binding energy per nucleon of a uranium-238 nucleus.

A. 2.08 MeV

B. 7.37 MeV

C. 448.59 MeV

D. 1754.77 MeV

A



B



C



D



(2011-PP)3.7

Which of the following is the function of the control rods in a fission reactor ?

- A. They slow down neutrons.
- B. They absorb neutrons.
- C. They absorb heat from the reactor.
- D. They transfer heat to the generator.

A

B

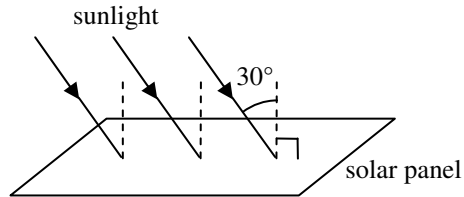
C

D



(2011-PP)3.8

Estimate the electrical power output of a 20 m^2 solar panel when it is illuminated with sunlight of intensity 1 kW m^{-2} at an angle of 30° to the vertical.



Given: efficiency of the solar panel = 12%

- A. 1200 W
- B. 1386 W
- C. 2078 W
- D. 2400 W

A



B



C



D



(2011-SP)B1. Which of these conclusions could **NOT** be deduced from Rutherford's scattering experiment ?

- (1) Alpha particles are helium nuclei.
- (2) There are discrete energy levels in an atom.
- (3) The positive charge in an atom is confined to a very small region.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Answer : _____

(2011-SP)B2. The equivalent wavelength of a photon of energy 10 eV is

A. 213 nm

B. 124 nm

C. 25.6 nm

D. 19.7 nm

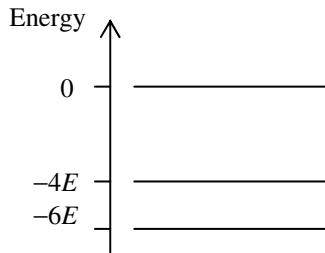
Answer : _____

(2011-SP)B3. The ionization potential of a hydrogen atom is 13.6 V. How much energy is required to excite an electron from the ground state to the first excited state in a hydrogen atom ?

- A. 10.2 eV
- B. 6.8 eV
- C. 3.4 eV
- D. 1.9 eV

Answer : _____

(2011-SP)B4.



The energy levels of a certain atom are as shown. Which of these may undergo an inelastic collision with the atom ?

- (1) an electron with kinetic energy $3E$
- (2) a photon with energy $2E$
- (3) a photon with energy $3E$

- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1), (2) and (3)

Answer : _____

(2011-SP)B5. In an experiment on the photoelectric effect, a beam of monochromatic light is directed onto a metal plate to liberate electrons. The velocity of the fastest photoelectrons emitted is

- A. directly proportional to the frequency of the incident light.
- B. directly proportional to the intensity of the incident light.
- C. independent of the nature of metal.
- D. independent of the intensity of the incident light.

Answer : _____

(2011-SP)^{B6.} The work function W of five metals are tabulated below.

Metal	Caesium	Barium	Calcium	Magnesium	Beryllium
$W/10^{-19} \text{ J}$	3.4	4.0	4.6	5.9	8.0

When monochromatic light of wavelength 400 nm is incident on each of the metals, how many of them would exhibit photoelectric emission ?

- A. 1
- B. 2
- C. 3
- D. 4

Answer : _____

(2011-SP)B7. Which of the following statements is/are correct ?

- (1) Photoelectric effect is an evidence that light possesses particle nature.
- (2) Electron diffraction suggests that electrons can behave like waves.
- (3) The line spectrum of atomic hydrogen suggests that the atom has discrete energy levels.

- A. (1) and (2) only
- B. (2) and (3) only
- C. (1) and (3) only
- D. (1), (2) and (3)

Answer : _____

(2011-SP)B8. Graphite is a conductor because of the ‘delocalization’ of electrons. Where are these delocalized electrons ?

- A. formed on the surface of graphite.
- B. formed within the carbon layers of graphite.
- C. formed homogeneously within graphite.
- D. formed in a ‘sea’ of positive ions.

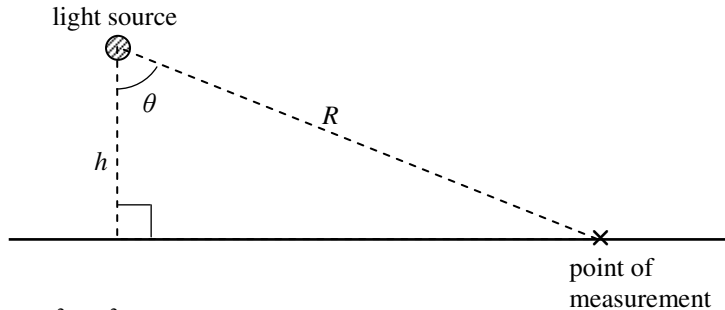
Answer : _____

(2011-SP)C1. The Coefficient of Performance (COP) of a heat pump is

- A. the ratio of energy absorbed from the cold reservoir to the energy rejected to the hot reservoir.
- B. the heat energy rejected to the hot reservoir per unit work input.
- C. the ratio between the total energy input to the useful work done.
- D. the ratio between the extra work input to the total energy input.

Answer : _____

(2011-SP)C2. In the figure, a light source is a perpendicular distance h above a horizontal surface. The amount of illuminance (unit: lux) of a point on the surface at a distance R from the source is directly proportional to



- A. $\cos^2\theta/R^2$
- B. $\cos^2\theta/h^2$
- C. $\cos^3\theta/h^2$
- D. $\cos^3\theta/R^2$

Answer : _____

(2011-SP)C3. If each fission of uranium-235 liberates 200 MeV of energy, how much uranium-235 must undergo fission per second to generate a power of 1000 MW ?

- A. 2.0×10^{-24} kg
- B. 2.2×10^{-10} kg
- C. 5.2×10^{-8} kg
- D. 1.2×10^{-5} kg

Answer : _____

(2011-SP)C4. Which of these is **NOT** an advantage of Battery Electric Vehicle ?

- A. zero emission
- B. low energy cost per kilometre covered
- C. long mileage range
- D. energy security by diversifying energy sources

Answer : _____

(2011-SP)C5. Which of these actions reduces the heat gained in the summer by buildings in Hong Kong ?

- A. Increase the OTTV values of the building envelope
- B. Apply solar films on windows to reduce solar heat gain
- C. Minimise internal heat gain from indoor activities
- D. Improve the air-tightness of the building envelope

Answer : _____

(2011-SP)C6. The solar constant is 1367 W m^{-2} (power per unit area from the Sun reaching the outer atmosphere) and the Earth-Sun distance is $1.50 \times 10^{11} \text{ m}$ (i.e. 1 AU), estimate the total radiation power of the Sun.

- A. $3.9 \times 10^{26} \text{ W}$
- B. $3.2 \times 10^{25} \text{ W}$
- C. $2.3 \times 10^{25} \text{ W}$
- D. $7.7 \times 10^{24} \text{ W}$

Answer : _____

(2011-SP)C7. In estimating the maximum power available from a wind turbine, what is assumed to be true ?

- (1) The density of air is constant.
- (2) The direction of wind relative to the orientation of the turbine is unchanged.
- (3) The area swept by the turbine is constant.

- A. (1) and (2) only
- B. (2) and (3) only
- C. (1) and (3) only
- D. (1), (2) and (3)

Answer : _____

(2011-SP)C8. A fuel cell cannot be classified as a Renewable Energy Source because

- A. it is a secondary energy source.
- B. its supply is limited.
- C. it is from fossil sources.
- D. the time scale for regeneration is too long.

Answer : _____

(2012) 2.1 From the classical point of view what are the limitations of Rutherford's model of the atom ?

- (1) Atoms would continuously emit electromagnetic radiation.
- (2) Atoms would be unstable and they would collapse eventually.
- (3) The atomic emission spectrum would be continuous instead of discrete.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A



B



C



D



(2012) 2.2 Which of the following statements about spectra is/are correct ?

- (1) A tungsten-filament lamp emits a continuous spectrum.
- (2) A line absorption spectrum can be obtained when a tungsten-filament lamp is viewed through some hydrogen gas.
- (3) The emission spectrum of hydrogen consists of dark lines on a bright background.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A

B

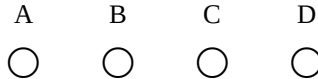
C

D

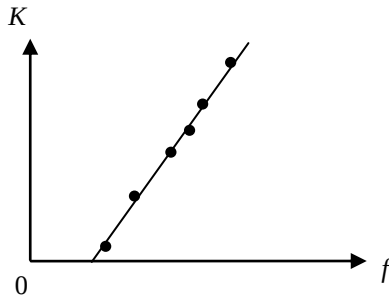


(2012) 2.3 Photons with energy 7 eV are incident on the cathode of a photocell. The maximum kinetic energy of the photoelectrons emitted is 4 eV. When photons of energy 4 eV are incident on the cathode, the stopping potential will be

- A. 0 V.
- B. 1 V.
- C. 2 V.
- D. 3 V.



- (2012) 2.4 In studying the photoelectric effect, a certain metal is illuminated by ultraviolet radiation of different frequency f and the maximum kinetic energy K of the photoelectrons emitted is measured. The graph is plotted as shown.



What would happen to the graph if ultraviolet radiation of higher intensity is shone on the same metal ?

slope of the graph

**intercept of the graph
on the horizontal axis**

- A. smaller
- B. larger
- C. unchanged
- D. unchanged

- unchanged
- unchanged
- unchanged
- smaller

A

B

C

D



(2012) 2.5 According to Bohr's model of the hydrogen atom, the ratio of the radius of the electron's orbit in the first excited state to that in the second excited state is

A. $1:2$.

B. $1:\sqrt{2}$.

C. $4:9$.

D. $2:3$.

A

B

C

D



(2012)

2.6 Which of the following has the shortest de Broglie wavelength ?

- A. A 60 kg person walking at 0.8 m s^{-1} .
- B. A bird of mass 0.3 kg flying at 20 m s^{-1} .
- C. A basketball of mass 0.6 kg moving at 12 m s^{-1} .
- D. A bullet of mass 0.05 kg moving at 800 m s^{-1} .

A

B

C

D



(2012)

2.7 Which of the following properties could explain the Lotus effect ?

A. water-attractive property

B. water-repelling property

C. wave-particle duality of matter

D. high electrical conductivity

A

B

C

D



(2012)

2.8 If substance is reduced in size to become particles of about 10 nm large, which of the following properties of these particles would differ from those of the substance in bulk form ?

- (1) optical
- (2) mechanical
- (3) electrical

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

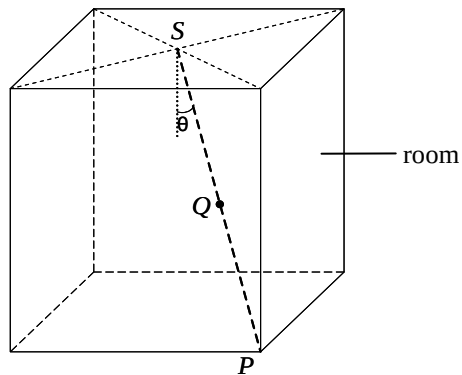
A	B	C	D
☐	☐	☐	☐

(2012) 3.1 Which of the following lamps has the greatest end-use energy efficiency ?

	luminous flux	power rating					
A.	750 lm	15 W	A	B	C	D	
B.	900 lm	30 W	☐	☐	☐	☐	
C.	750 lm	60 W					
D.	600 lm	90 W					

(2012)

- 3.2 Below is a diagram of a room, the illuminance at corner P is E under the illumination of a point light source S as shown. Neglect the reflections from the room surfaces.



The illuminance at point Q midway between P and S is

A. $\frac{\sqrt{2}E}{\cos \theta}$.

B. $2E \cos \theta$.

C. $4E$.

D. $8E$.

A



B



C



D



(2012)

3.3 Which sequence best describes the energy conversion in an electric vehicle's regenerative braking system during braking ?

- A. kinetic energy $\rightarrow$ electrical energy
- B. kinetic energy $\rightarrow$ chemical energy
- C. kinetic energy $\rightarrow$ chemical energy $\rightarrow$ electrical energy
- D. kinetic energy $\rightarrow$ electrical energy $\rightarrow$ chemical energy

A



B



C



D

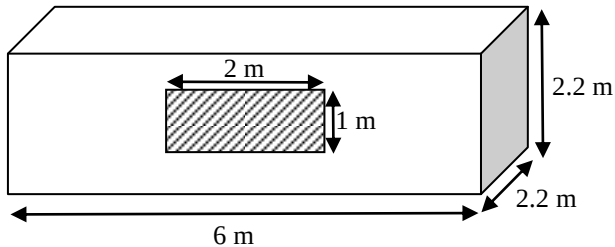


(2012)

- 3.4 A container is modified into an office as shown. A window of $1\text{ m} \times 2\text{ m}$ is installed on the front side of the container. Apart from the bottom, assume that all the five surfaces of the container are under sunlight. On all the five surfaces, the equivalent temperature difference between the interior and the exterior of the container is 7°C .

Given: U-value of metallic material of the container = $26.2\text{ W m}^{-2}\text{ K}^{-1}$

U-value of glass of the window = $1.8\text{ W m}^{-2}\text{ K}^{-1}$



Estimate the Overall Thermal Transfer value (OTTV) in W m^{-2} of the container office.

- A. 25.2
B. 26.2
C. 176.5
D. 183.4

A

B

C

D



(2012) 3.5 Estimate the time required to cool an insulated room of interior volume 29.0 m^3 from 37°C to 24°C by an air conditioner of cooling capacity 2.2 kW .

Given: density of air = 1.2 kg m^{-3}

specific heat capacity of air = $1000 \text{ J kg}^{-1} \text{ K}^{-1}$

A. 171 s

B. 206 s

C. 380 s

D. 586 s

A

B

C

D



(2012) 3.6 The blades of a wind turbine are 5 m long, which are set to rotate when wind blows normally at 12 m s^{-1} . The overall efficiency of the wind turbine is 25%. Estimate the number of wind turbines required to generate an electrical power output of 1 MW. Given: density of air = 1.2 kg m^{-3}

- A. 12
- B. 49
- C. 122
- D. 196

A	B	C	D
☐	☐	☐	☐

(2012) 3.7 The difference in water levels of a hydroelectric power plant is 50 m. Water passes through the turbine of the power plant at the lower water level at a rate of 3000 kg s^{-1} . Estimate the efficiency of the turbine if its output power is 1 MW. ($g = 9.81 \text{ m s}^{-2}$)

- A. 32%
- B. 60%
- C. 68%
- D. 75%

A

B

C

D



(2012) 3.8 What would happen if the moderator of a nuclear fission reactor fails to function ?

- A. The chain reaction might stop eventually.
- B. Neutrons cannot be absorbed by the moderator.
- C. Heat cannot be transferred to the steam generator.
- D. The fuel rods might melt down.

A

B

C

D



(2013) 2.1 In an α -particle scattering experiment, the electrons of the atoms have negligible effects on the path of the incident α -particles. The most probable reason is that

- A. electrons are so small that α -particles do not collide with them.
- B. electrons are uniformly distributed inside an atom, thus the resultant force on an α -particle is zero.
- C. there is no electrical interaction between electrons and α -particles.
- D. the kinetic energy change of α -particles colliding with electrons is negligible.

A	B	C	D
☐	☐	☐	☐

(2013) 2.2 According to classical electromagnetic theory, what deductions about Rutherford's atomic model can be made ?

- A. Atoms are stable and atomic spectra are continuous spectra.
- B. Atoms are stable and atomic spectra are line spectra.
- C. Atoms are unstable and atomic spectra are continuous spectra.
- D. Atoms are unstable and atomic spectra are line spectra.

A

☐

B

☐

C

☐

D

☐

(2013) 2.3 Which of the following spectra is/are continuous ?

- (1) the spectrum from a burning candle
- (2) the spectrum from an incandescent lamp
- (3) the spectrum from a gas discharge tube

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2013) 2.4 The energy level of an electron in a hydrogen atom is given by $E_n = -\frac{E_0}{n^2}$, where E_0 is a constant and $n = 1, 2, 3, \dots$. What is the maximum wavelength of a photon that can ionize a hydrogen atom in its first excited state ? (h = Planck constant, c = speed of light in a vacuum)

A. $\frac{3hc}{4E_0}$

B. $\frac{hc}{E_0}$

C. $\frac{4hc}{3E_0}$

D. $\frac{4hc}{E_0}$

A	B	C	D
☐	☐	☐	☐

(2013) 2.5 When an electron in a hydrogen atom falls from an excited state to the ground state, which of the following forms of radiation will **NOT** be emitted ?

- | | | | | | |
|----|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | γ -radiation | A | B | C | D |
| B. | ultra-violet radiation | ☐ | ☐ | ☐ | ☐ |
| C. | infra-red radiation | | | | |
| D. | visible light | | | | |

Figure (1)

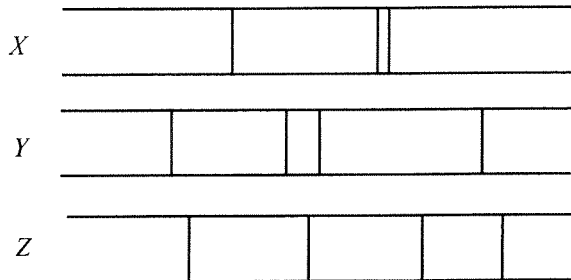


Figure (2)



Figure (1) shows the line spectra of three elements X , Y and Z while Figure (2) shows the line spectrum of a certain mineral substance. From spectral analysis, which element must be **ABSENT** from the mineral substance ?

- | | | | | | |
|----|--|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | X | A | B | C | D |
| B. | Y | ☐ | ☐ | ☐ | ☐ |
| C. | Z | | | | |
| D. | All three elements are present in the mineral substance. | | | | |

(2013) 2.7 The minimum resolvable length of a typical transmission electron microscope (TEM) is about 0.2 nm. If a particle has the same charge of an electron and its mass is four times that of an electron, and a beam of such particles is accelerated through the same voltage in a TEM, the minimum resolvable length becomes

A. 0.05 nm.

B. 0.1 nm.

C. 0.4 nm.

D. 0.8 nm.

A

B

C

D



(2013) 2.8 A cube with 1 mm per side is divided into nano-scale cubes, each side measuring 1 nm. How many times has the total surface area of the cube been increased ?

- A. 10^6
- B. 10^8
- C. 10^{10}
- D. 10^{12}

A



B



C



D

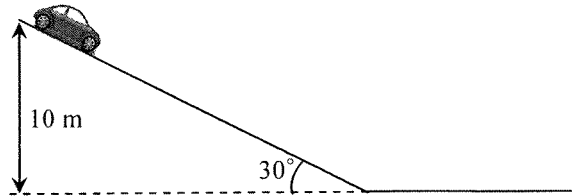


(2013) 3.1 LED usually produces monochromatic light because

- A. the difference in energy levels of the p-layer and n-layer in an LED is fixed.
- B. its surface is coated with only one type of fluorescent material.
- C. the operating temperature of an LED is fixed.
- D. current is only allowed to flow through an LED in one direction.

A	B	C	D
☐	☐	☐	☐

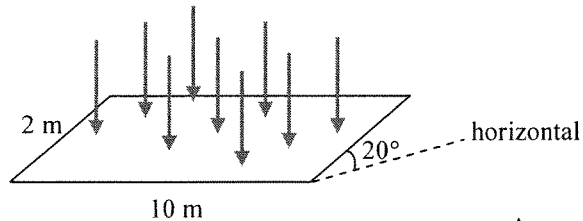
- (2013) 3.2 A car of mass 1000 kg has a regenerative braking system that can convert its kinetic energy into chemical energy which is stored in a battery of that system. How much energy can be stored in the battery after the car has gone downhill at a constant speed by a vertical distance of 10 m ? The overall efficiency of the regenerative braking system is 30%. ($g = 9.81 \text{ m s}^{-2}$)



- A. 14.7 kJ
- B. 29.4 kJ
- C. 49.1 kJ
- D. 98.1 kJ

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

- (2013) 3.3 Assume that 1000 W m^{-2} of solar power reaches a certain place vertically on the Earth's surface. What is the power delivered to a solar panel of width 2 m and length 10 m inclined at 20° to the horizontal at that place ?



- A. 6840 W
- B. 7280 W
- C. 18800 W
- D. 20000 W

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2013) 3.4 A wind turbine is used to drive a pump which pumps water up to a storage reservoir. The blade-length of the wind turbine is 10 m while the average wind speed is 5 m s^{-1} . How much water can be pumped up to the reservoir in 8 hours if the overall efficiency of the system is 20% ? Assume that the average gain in gravitational potential energy by water is 981 J kg^{-1} . Given: density of air = 1.23 kg m^{-3}

- A. 39.4 kg
- B. $2.84 \times 10^4 \text{ kg}$
- C. $1.15 \times 10^5 \text{ kg}$
- D. $1.42 \times 10^5 \text{ kg}$

A	B	C	D
☐	☐	☐	☐

(2013) 3.5 A closed foam box of dimensions $0.5 \text{ m} \times 0.3 \text{ m} \times 0.4 \text{ m}$ and thickness 1 cm contains some melting ice. Estimate the rate of heat conducting from the surroundings into the box if the room temperature is 28°C .
Given: thermal conductivity of the foam is $0.03 \text{ W m}^{-1} ^\circ\text{C}^{-1}$

- A. 39.5 W
- B. 79 W
- C. 3950 W
- D. 7900 W

A	B	C	D
☐	☐	☐	☐

(2013) 3.6 The Overall Thermal Transfer Value (OTTV) of a building can be reduced by making its glass windows smaller because

- (1) glass has a much higher thermal conductivity than concrete.
- (2) heat can be transferred by convection if windows are open.
- (3) glass allows heat transfer by radiation.

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

A

B

C

D



ENERGY LABEL 能源標籤	
Brand 牌子	ABC 某某牌
Model 型號	HK1234
Annual Energy Consumption due to Standby Loss * kWh/yr 待機耗電量 (每年度) Standby loss is the energy lost when the water heater is idle. Actual consumption depends on the amount of hot water used. Based on using 100 L of 180°C water. 待機耗電量為熱水器在閒用時消耗的電量。實際耗電量需視乎熱水用量。此值按每年熱水總用量約為1800 L計算。	85
Energy Efficiency Grade* 能源效率級別 Among the five grades, Grade 1 is the most energy efficient. 於五級中，第一級最省電。	2
Water Heater Category * 熱水爐類別	1
Rated Capacity (L) 容量 (升)	40
Heating Time* (min) 加熱時間 (分) (15°C → 65°C)	75
EEL Registration Number 能源標籤登記號碼	1100-0001
* The data are provided according to the Hong Kong Energy Efficiency Labeling Scheme for Household Electric Storage Water Heaters administered by the Electrical and Mechanical Services Department (EMSD), Government of the Hong Kong Special Administrative Region. The registration records can be found at the EMSD website at www.emsd.gov.hk. 資料根據香港強制性綠標政府機構工程署行政的香港家用電式儲熱式熱水器能源標籤計劃提供。 有關註冊記錄可查閱網址 www.emsd.gov.hk。	

Rated Capacity : 40 L

Heating time (15 °C → 65 °C): 75 min.

Referring to the above energy label of a storage type water heater, estimate the effective output power of the heater when it is filled to maximum capacity with 40 L (1 L = 1000 cm³) of water at 15 °C.

Given: specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹, density of water = 1000 kg m⁻³

- A. 1870 W
B. 2430 W
C. 85000 W
D. 112000 W

- A B C D
○ ○ ○ ○

(2013) 3.8 When four hydrogen nuclei, each of mass 1.007825 u, are fused together, one nucleus of element Q of mass 4.002603 u is formed. Which of the following statements is/are correct ?

- (1) Hydrogen nuclei need to have extremely high kinetic energy to start the fusion process.
- (2) Energy released in the process is about 26.7 MeV.
- (3) Element Q is radioactive.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2014) 2.1 Which of the following statements about Rutherford's atomic model is/are correct ?

- (1) All the positive charges and nearly all of the atom's mass are concentrated in its tiny central nucleus.
- (2) Negatively charged electrons orbit around the nucleus.
- (3) The electrons can only move in certain orbits each having a definite energy and they do not radiate while accelerating.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A

B

C

D



(2014) 2.2 There are dark lines in the spectrum of sunlight. Which of the following statements are correct ?

- (1) They are due to the absorption of certain wavelengths of light by the atoms in the Sun's atmosphere.
- (2) Light absorbed by the atoms in the Sun's atmosphere is then re-emitted in all directions.
- (3) The kinds of atoms present in the Sun's atmosphere can be deduced by the characteristics of the dark lines.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2014) 2.3 A hydrogen atom in the ground state absorbs a photon of wavelength λ so that it is excited from quantum number $n = 1$ to $n = 3$. The ionization energy, in eV, of a hydrogen atom in the ground state is given by

A. $\frac{3hc}{2\lambda}$.

B. $\frac{2hc}{3\lambda}$.

C. $\frac{9hc}{8\lambda}$.

D. $\frac{8hc}{9\lambda}$.

A



B



C

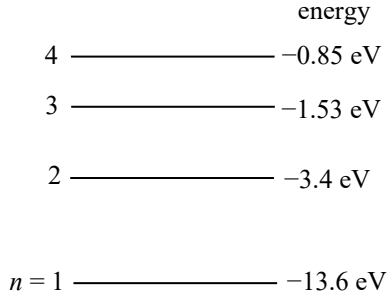


D



(2014)

2.4



The lowest four energy levels of a hydrogen atom are shown above. Which statement about a hydrogen atom in the ground state is **INCORRECT** ?

- A. The collision between an electron of kinetic energy 10 eV and the atom is perfectly elastic.
- B. The collision between an electron of kinetic energy 12 eV and the atom can be inelastic.
- C. A photon of energy 12 eV can excite the atom to the first excited state.
- D. A photon of energy 14 eV can ionize the atom.

A

B

C

D



(2014) 2.5 When photons each of energy 3.41 eV are incident on a metal surface, the maximum kinetic energy of the photoelectrons emitted is 0.54 eV. What is the threshold frequency of this metal ?

A. 4.33×10^{33} Hz

B. 9.53×10^{14} Hz

C. 8.23×10^{14} Hz

D. 6.93×10^{14} Hz

A

B

C

D



- (2014) 2.6 A beam of light of frequency f falls on the cathode of a photocell so that photoelectrons are emitted. If the light beam is replaced by another one with the same intensity but having a frequency of $2f$, how would each of the following physical quantities change? Assume that each incident photon can emit one photoelectron.

V_s : stopping potential

I : magnitude of the saturation photoelectric current

	V_s	I				
A.	increases	increases	A	B	C	D
B.	increases	decreases	☐	☐	☐	☐
C.	remains unchanged	decreases				
D.	decreases	increases				

(2014) 2.7 The de Broglie wavelength of object X is shorter than that of object Y . Which of the following deductions must be correct ?

- (1) X has a higher speed than Y .
- (2) X has a greater momentum than Y .
- (3) X has greater kinetic energy than Y .

A. (2) only

B. (1) and (2) only

C. (2) and (3) only

D. (1), (2) and (3)

A

B

C

D



(2014) 2.8 Which statements about nanotechnology are correct ?

- (1) Glass can be made self-cleaning by coating it with titanium oxide in nanoscale.
- (2) Carbon nanotubes and diamonds have the same spatial arrangement of carbon atoms.
- (3) When gold is reduced to nanoscale, its melting point would be different from that of its bulk form.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

A

B

C

D



(2014) 3.1 For monochromatic light sources, which of the following pairs appear(s) to be equally bright to our eyes ?

- (1) 1 lumen of red light and 1 lumen of green light
- (2) red light of 1 watt and green light of 1 watt
- (3) red light of 1 watt and 1 lumen of red light

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

A

B

C

D



- (2014) 3.2 The energy labels for air-conditioners show the annual energy consumption (based on 1200 hours of operation per year) and the cooling capacity. Which air-conditioner below is the most energy efficient ?

	annual energy consumption / kW h	cooling capacity / kW				
A.	672	2.44	A	B	C	D
B.	684	2.58	☐	☐	☐	☐
C.	696	2.89				
D.	750	2.63				

(2014) 3.3 Which of the following building materials with thicknesses listed below give the best heat insulation ?

	material	thermal conductivity / $\text{W m}^{-1} \text{K}^{-1}$	thickness / m				
A.	concrete	0.50	0.20	A	B	C	D
B.	wood	0.15	0.05	☐	☐	☐	☐
C.	glass	1.00	0.04				
D.	plaster	0.24	0.10				

(2014) 3.4 Glass with low-emissivity coating can reduce heat transfer into a building by reflecting mainly

A. ultra-violet radiation.

B. visible light.

C. infra-red radiation.

D. microwaves.

A

B

C

D



(2014) 3.5 A wind turbine generator experiences wind blowing normal to it with variable speed such that the wind speed is 1 m s^{-1} for the first two minutes and 2 m s^{-1} for the third minute. What is its average power output, in W, for this period of 3 minutes if the overall efficiency of the generator is 30% and the length of each blade is 20 m ? Given : ρ = density of air in kg m^{-3} .

- A. $180 \pi \rho$
- B. $200 \pi \rho$
- C. $600 \pi \rho$
- D. $667 \pi \rho$

A	B	C	D
☐	☐	☐	☐

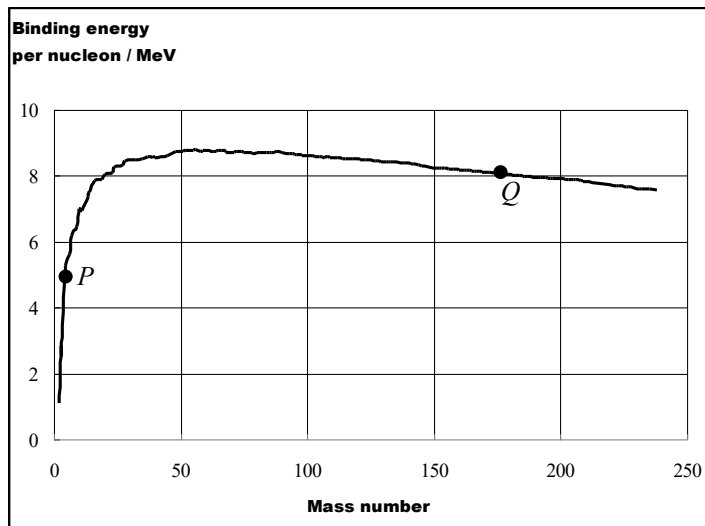
(2014) 3.6 Which of the following statements about hybrid vehicles is/are correct ?

- (1) The battery of a hybrid vehicle needs to be recharged by an external electric source before the vehicle can run.
- (2) The power of the internal combustion engine of a hybrid vehicle is smaller than that of a conventional petrol vehicle of the same weight and performance.
- (3) The primary energy source of a hybrid vehicle is 100% petrol.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2014) 3.7 The curve below shows the binding energy per nucleon of nuclides of different mass numbers.



Which of the following statements is/are correct ?

- (1) Nuclei of P can release energy by nuclear fusion.
- (2) Nuclei of Q can release energy by nuclear fission.
- (3) Nuclei of P are more stable than nuclei of Q .

- A. (2) only
- B. (1) and (2) only
- C. (1) and (3) only
- D. (1), (2) and (3)

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2014) 3.8 Under normal operation, which of the following statements about a pressurized water reactor (PWR) of a nuclear power plant is/are correct ?

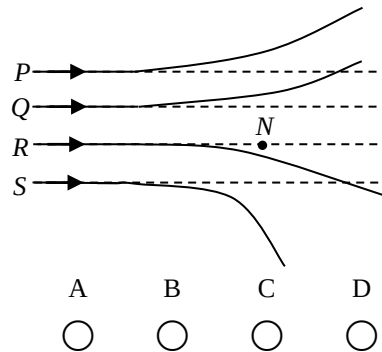
- (1) The coolant which carries energy away from the reactor is radioactive.
- (2) The steam that drives the turbine is radioactive.
- (3) The cooling water discharged into the sea from the nuclear power plant contains some radioactive substances of the reactor.

- | | | | | | |
|----|------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | (1) only | A | B | C | D |
| B. | (3) only | ☐ | ☐ | ☐ | ☐ |
| C. | (1) and (2) only | | | | |
| D. | (2) and (3) only | | | | |

(2015)

- 2.1 A beam of α -particles with the same initial kinetic energy are scattered by a heavy nucleus N . In the figure, if P is a possible path for one of the α -particles, which of the paths, Q , R and S , is/are possible for these α -particles ?

- A. Q and R only
- B. R and S only
- C. Q only
- D. S only



(2015) 2.2 Which of the following provides experimental evidence for **discrete energy levels** in atoms ?

- (1) the spectrum of a sodium discharge tube
- (2) the spectrum of a tungsten filament lamp
- (3) the diffraction of electrons by atomic spacing in crystals

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

A

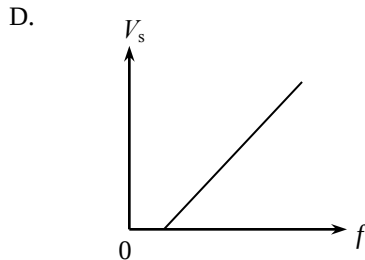
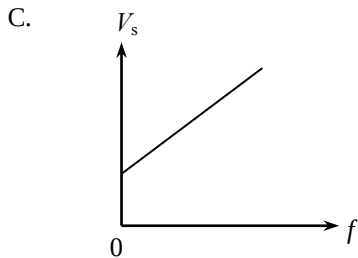
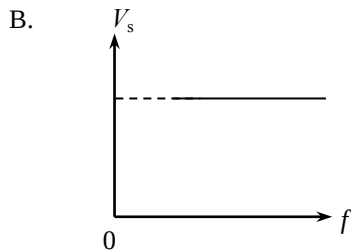
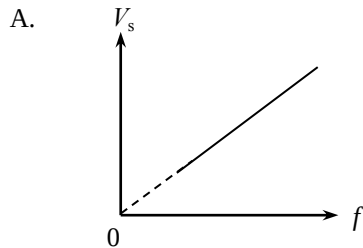
B

C

D



- (2015) 2.3 In a photoelectric experiment, monochromatic light of frequency f is incident upon a metal surface and the stopping potential for the photoelectrons emitted is V_s . If the frequency f is changed, how would V_s vary with f ?



A



B



C



D

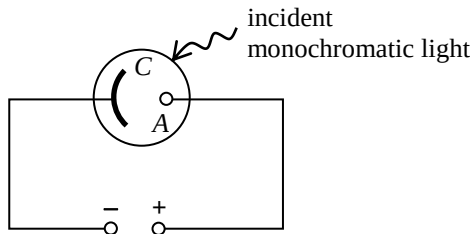


(2015)

- 2.4 A spy aircraft is cruising at a height of 10 km above the Earth's surface. It is equipped with a camera having an objective lens of aperture 10 cm. Estimate the minimum separation of two small objects on the Earth's surface to be distinguished by this camera. Assume that both objects emit light of wavelength 500 nm.

- A. 0.05 m
- B. 0.061 m
- C. 0.10 m
- D. 0.122 m





A photocell is connected to a d.c. source as shown. Monochromatic light is incident on cathode C of the photocell so that photoelectrons are emitted. The maximum kinetic energy of photoelectrons reaching anode A depends on

- (1) the kind of metal making the cathode surface.
- (2) the voltage of the d.c. source.
- (3) the intensity of the monochromatic light used.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2015) 2.6 Aurora Borealis (Northern lights) are often observed in skies at high latitudes. When energetic electrons from outer space collide with the oxygen atoms in the upper atmosphere, the oxygen atoms are excited. The subsequent emission of light is usually green light of wavelength 558 nm. The minimum speed of these energetic electrons is of order of magnitude

A. 10^2 m s^{-1} .

B. 10^4 m s^{-1} .

C. 10^6 m s^{-1} .

D. 10^8 m s^{-1} .

A

B

C

D



(2015)

2.7 Which of the following can increase the resolving power of a transmission electron microscope (TEM) ?

- (1) increasing the anode voltage in the electron gun
- (2) decreasing the aperture of the magnetic objective lens
- (3) increasing the separation between the magnetic projection lens and the fluorescent screen

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

A

B

C

D

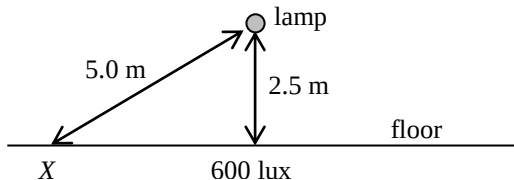


(2015) 2.8 Zinc oxide (ZnO) is used in some suntan lotion. Which of the following statements is/are correct ?

- (1) Nano-sized ZnO can block ultra-violet radiation while bulk-sized ZnO cannot.
- (2) Nano-sized ZnO is more effective in reflecting visible light than bulk-sized ZnO.
- (3) Suntan lotion with nano-sized ZnO appears transparent when applied to the skin.

- | | | | | | |
|----|------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | (1) only | A | B | C | D |
| B. | (3) only | ☐ | ☐ | ☐ | ☐ |
| C. | (1) and (2) only | | | | |
| D. | (2) and (3) only | | | | |

- (2015) 3.1 A lamp is fixed on the ceiling of a room as the only light source. The illuminance on the floor directly under the lamp is 600 lux. Assuming that the lamp emits light uniformly in all directions, what is the illuminance on the floor around point X as indicated in the diagram? Neglect any reflections of the walls and the ceiling.

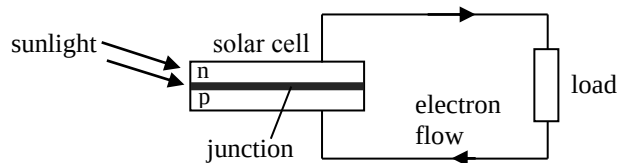


- A. 75 lux
B. 130 lux
C. 150 lux
D. 300 lux

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2015)

- 3.2 The schematic diagram below shows a solar cell under sunlight. Incident light photons reach the cell's junction of p-type and n-type semiconductor layers. Which of the following sequences correctly explains how electrical energy can be supplied to the load ?



- (1) The electric field developed within the junction draws free electrons and holes to the n-type and p-type layers respectively.
- (2) Free electrons flow to the p-type layer through the load and rejoin the holes there.
- (3) Incident light photons knock out electrons from atoms in the junction such that free electrons result, leaving the holes behind.

- A. (1) → (2) → (3)
B. (1) → (3) → (2)
C. (3) → (2) → (1)
D. (3) → (1) → (2)

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2015)

3.3 A satellite is powered by a solar panel of an area of 100 m^2 and having a conversion efficiency of 15%. Sunlight falls from a direction 30° to the normal of the panel. The solar constant is 1370 W m^{-2} . Estimate the electrical power output of this solar panel.

- A. 10.3 kW
- B. 17.8 kW
- C. 20.6 kW
- D. 58.2 kW

A	B	C	D
☐	☐	☐	☐

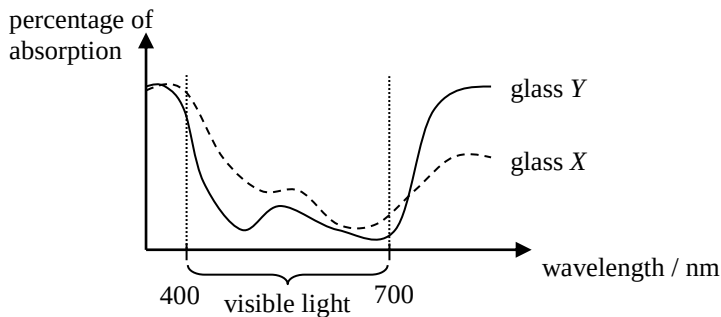
- (2015) 3.4 A coal-fired power plant generates and transmits electrical energy to consumers a long distance away. When a consumer connects an incandescent lamp to the mains supply, the following data illustrates what happens per 1000 J of energy in coal converted to electrical energy which is then supplied to the lamp.

Loss during generation process in the power plant	600 J
Loss during transmission before reaching the lamp	100 J
Heat generated by the lamp	250 J
Visible light generated by the lamp	50 J

What is the **end-use energy efficiency** of this incandescent lamp ?

- | | | | | | |
|----|--------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | 5 % | A | B | C | D |
| B. | 12.5 % | ☐ | ☐ | ☐ | ☐ |
| C. | 16.7 % | | | | |
| D. | 30 % | | | | |

- (2015) 3.5 The following graph shows the percentage of absorption of different electromagnetic radiations when passing through two types of glasses *X* and *Y*. One of them is to be chosen for windows of buildings in Hong Kong. With the consideration of energy efficiency of buildings, which statement is correct ?

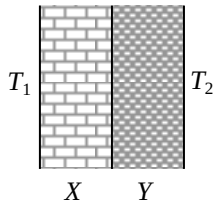


- A. *X* should be chosen as it is better than *Y* in reducing the need of air conditioning and the amount of lighting.
- B. *Y* should be chosen as it is better than *X* in reducing the need of air conditioning and the amount of lighting.
- C. *X* should be chosen as it can greatly reduce the need of air conditioning although it needs a bit more amount of lighting than *Y*.
- D. *Y* should be chosen as it can greatly reduce the need of air conditioning although it needs a bit more amount of lighting than *X*.

A	B	C	D
☐	☐	☐	☐

(2015)

- 3.6 The diagram shows a wall composed of two layers X and Y of equal thicknesses. The thermal conductivity of the material of X is higher than that of Y . The two sides of the wall are maintained at different temperatures T_1 and T_2 . Which of the following statements is/are correct ?

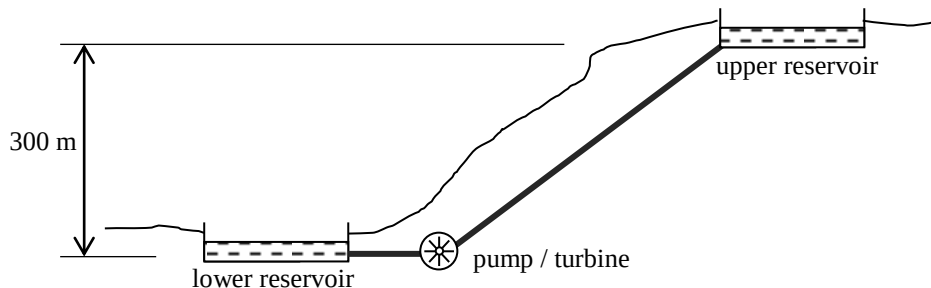


- (1) The thermal transmittance of layer X is higher than that of layer Y .
- (2) The energy flowing through layer X per second is greater than that through layer Y .
- (3) The temperature drop across layer X is smaller than that across layer Y .

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

- (2015) 3.7 A power plant is always running at its maximum output power of 1800 MW. However, as the demand of electricity fluctuates greatly within a day, a hydroelectric storage system is therefore designed to increase the output in the high demand period. During the 14 hours of low demand period in a day, the average demand is only 600 MW and the excess output is used to pump water from the lower reservoir to the upper one. When the demand is higher than 1800 MW, the water returns to the lower reservoir through a turbine to generate electricity.



Assume that the hydroelectric storage system is 100% efficient, find the minimum capacity (in kg of water) of the upper reservoir required. ($g = 9.81 \text{ m s}^{-2}$)

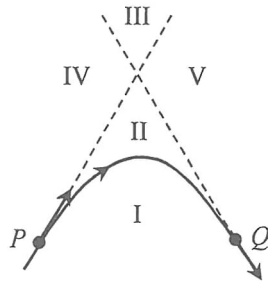
- A. $1.03 \times 10^{10} \text{ kg}$
- B. $1.47 \times 10^{10} \text{ kg}$
- C. $2.06 \times 10^{10} \text{ kg}$
- D. $3.08 \times 10^{10} \text{ kg}$

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2015) 3.8 What is the function of the **moderator** in a nuclear fission reactor ?

- A. It slows down neutrons and this helps increase the rate of nuclear fission.
- B. It slows down neutrons and this helps reduce the rate of nuclear fission.
- C. It absorbs neutrons and this helps reduce the rate of nuclear fission.
- D. It generates neutrons for nuclear fission.

A	B	C	D
☐	☐	☐	☐



In the above figure, the solid line is the trajectory of an α -particle scattered by a gold nucleus (not shown in figure). The dotted lines are tangents to the trajectory at points P and Q . The two dotted lines together with the trajectory divide the plane into five regions (I – V). In which region(s) can the gold nucleus be situated ?

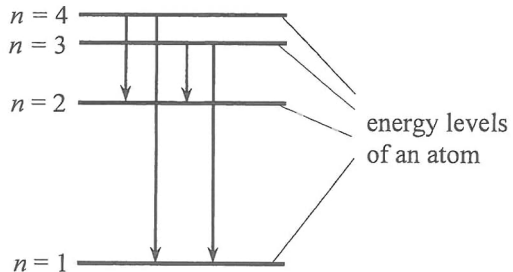
- A. I
- B. II
- C. III
- D. IV or V

A	B	C	D
☐	☐	☐	☐

(2016) 2.2 Which statements about *wave-particle duality* are correct ?

- (1) Interference of light is evidence that light behaves as a wave.
- (2) Photoelectric effect is evidence that light behaves as a particle.
- (3) Electron diffraction by a crystal shows that electrons behave as a wave.

A.	(1) and (2) only	A	B	C	D
B.	(1) and (3) only	☐	☐	☐	☐
C.	(2) and (3) only				
D.	(1), (2) and (3)				



The above figure shows four energy levels of an atom drawn approximately to scale. Which emission spectrum below best corresponds to the four electron transitions indicated?

increasing frequency $\rightarrow$

A.		A	B	C	D
B.		☐	☐	☐	☐
C.					
D.					

(2016) 2.4 Which of the following electron transitions between energy levels in a hydrogen atom will emit electromagnetic radiation of the longest wavelength ?

A. $n = 2$ to $n = 1$

B. $n = 3$ to $n = 2$

C. $n = 4$ to $n = 2$

D. $n = 5$ to $n = 2$

A

B

C

D



(2016)

2.5 If the de Broglie waves associated with a proton and an α -particle have the same wavelength, what is the ratio of the kinetic energy of the proton to that of the α -particle ?

A. 1 : 4

B. 4 : 1

C. 1 : 2

D. 2 : 1

A



B



C



D



(2016) 2.6 The minimum angular separation between two points which can be resolved by the human eye (pupil diameter 4 mm) under normal lighting has the order of magnitude of

A. 10^{-1} rad.

B. 10^{-2} rad.

C. 10^{-3} rad.

D. 10^{-4} rad.

A



B



C



D



(2016)

2.7 Transmission electron microscope (TEM) is used to observe structures of nano-scale instead of an optical microscope. This is because electron wave compared to visible light can have

- A. shorter wavelength so that its diffraction is less significant.
- B. shorter wavelength so that its diffraction is more significant.
- C. longer wavelength so that its diffraction is less significant.
- D. longer wavelength so that its diffraction is more significant.

A

☐

B

☐

C

☐

D

☐

(2016) 2.8 Which of the following applications in nanotechnology utilize(s) **Lotus effect** ?

- (1) Water-repelling fabric used in swimming suits is manufactured by nano-coating.
- (2) Glass is made self-cleaning by coating it with a water-attracting material in nanoscale.
- (3) Nano-sized zinc oxide is added to fabric as a photocatalyst for protection from dirt.

- | | | | | | |
|----|------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | (1) only | A | B | C | D |
| B. | (1) and (2) only | ☐ | ☐ | ☐ | ☐ |
| C. | (1) and (3) only | | | | |
| D. | (2) and (3) only | | | | |

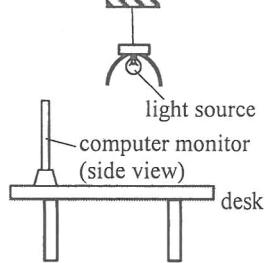
(2016)

3.1 In a computer room, there is a light source installed directly above each desk such that a concave reflector encloses most of the source as shown. The purpose(s) of such arrangement is/are to

- (1) maximize the amount of light reaching the desk's surface.
- (2) reduce the amount of light reaching the computer monitor so that glare can be minimized.
- (3) increase the luminous flux of the light source.

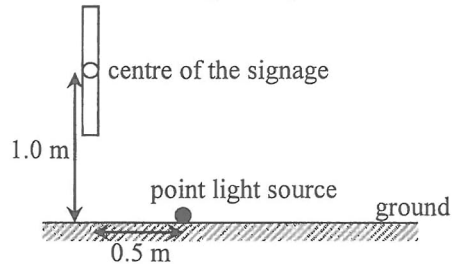
- A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

- A B C D
- ☐ ☐ ☐ ☐



(2016)

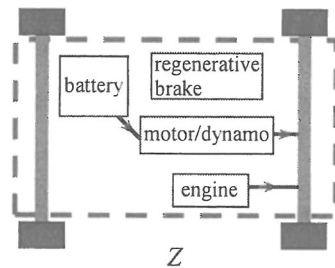
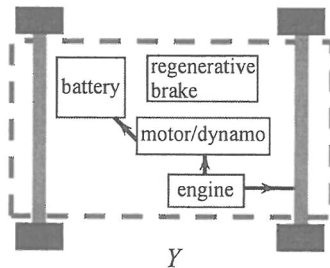
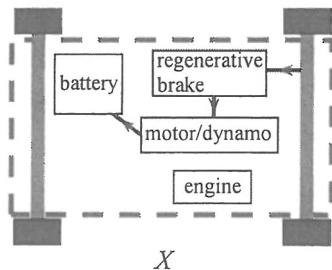
- 3.2 The signage below is to be illuminated by a point light source from the ground as shown. What is the luminous flux of the source required for producing an illuminance of 200 lux at the centre of the signage ? Assume that the source emits light uniformly in all directions and neglect any reflection from the ground.



- A. 7025 lm
B. 3512 lm
C. 3142 lm
D. 560 lm

A	B	C	D
☐	☐	☐	☐

(2016) 3.3 Figures *X*, *Y* and *Z* below show the energy flow of a hybrid car in three different situations.



Which of the following correctly matches the figures with these situations ?

	braking	accelerating with full power	cruising at a constant speed
A.	<i>X</i>	<i>Y</i>	<i>Z</i>
B.	<i>Y</i>	<i>X</i>	<i>Z</i>
C.	<i>X</i>	<i>Z</i>	<i>Y</i>
D.	<i>Y</i>	<i>Z</i>	<i>X</i>

A	B	C	D
☐	☐	☐	☐

(2016) 3.4 Which statement about a solar cell is **INCORRECT** ?

- A. When sunlight shines on a solar cell, some electrons in the semiconductor are excited to be free electrons.
- B. When a solar cell delivers power, current only flows in the interface at the p-n junction.
- C. The output voltage of a solar cell remains more or less unchanged when the incident light intensity increases.
- D. The efficiency of typical solar cells is about 10% to 20%.

A

☐

B

☐

C

☐

D

☐

(2016) 3.5 A room is cooled down by an air-conditioner and maintained a steady temperature which is ΔT below the temperature outside. Which of the factors below would affect ΔT ?

- (1) thermal conductivity of the wall material of the room
- (2) cooling capacity of the air-conditioner
- (3) specific heat capacity of air

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2016) 3.6 Which of the following expressions take(s) the unit watt (W) ?

- (1) luminous flux $\times$ area
- (2) solar constant $\times$ area
- (3) thermal transmittance $\times$ area $\times$ temperature difference

- A. (2) only
- B. (1) and (2) only
- C. (1) and (3) only
- D. (2) and (3) only

A



B



C



D

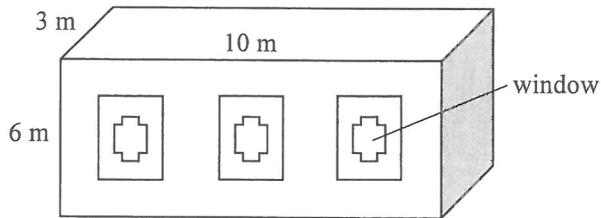


(2016) 3.7 An air-conditioner can remove 1 J of heat from a room to the exterior for every 0.5 J it consumes. Estimate the total thermal energy released to the exterior when the air-conditioner removes 1500 J of heat from the room.

- A. 750 J
- B. 1000 J
- C. 2250 J
- D. 4500 J

A	B	C	D
☐	☐	☐	☐

- (2016) 3.8 The Overall Thermal Transfer Value (OTTV) of the house shown is 25 W m^{-2} and the rate of heat generated due to human activities inside is 2000 W . Which cooling capacity below for an air-conditioning system is the **most appropriate** choice for the house ?



- A. 2 kW
- B. 5 kW
- C. 10 kW
- D. 15 kW

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2017) 2.1 Which of the following statements are correct according to Rutherford's atomic model ?

- (1) Almost all the mass of an atom are concentrated at the nucleus.
- (2) Almost all the charges of an atom are concentrated at the nucleus.
- (3) Electrons orbit around the nucleus.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A

B

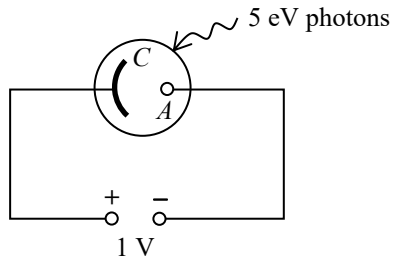
C

D



(2017)

2.2



A photocell is connected to a 1 V d.c. source as shown. A monochromatic light beam with each photon of energy 5 eV is incident on cathode C of the photocell so that photoelectrons are emitted. If the work function of cathode C is 2 eV, what is the maximum kinetic energy of the photoelectrons reaching anode A ?

- A. 2 eV
- B. 3 eV
- C. 4 eV
- D. 6 eV

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2017)

2.3 When monochromatic light of wavelengths λ and $\frac{3}{4}\lambda$ are incident on the cathode surface of a photocell separately, the stopping potentials are in the ratio of 1 : 2. What is the longest wavelength of monochromatic light that can cause photoelectrons to be emitted from the photocell ?

A. λ

B. $\frac{4}{3}\lambda$

C. $\frac{3}{2}\lambda$

D. $\frac{5}{3}\lambda$

A

B

C

D



(2017)

2.4 A parallel beam of yellow light from a sodium discharge tube is directed to a glass tube filled with sodium vapour. Which of the following would happen after the sodium vapour absorbs the yellow light ?

- A. No more yellow light can be seen.
- B. The sodium vapour emits yellow light in the direction of the incident beam.
- C. The sodium vapour emits yellow light in all directions.
- D. The sodium vapour emits white light in all directions.

A

B

C

D



(2017)

2.5 A beam of 8 keV electrons is directed towards a crystal to observe the diffraction of electrons. What is the de Broglie wavelength of a 8 keV electron ?

A. $4.34 \times 10^{-10} \text{ m}$

B. $1.37 \times 10^{-11} \text{ m}$

C. $1.74 \times 10^{-19} \text{ m}$

D. $5.49 \times 10^{-21} \text{ m}$

A

B

C

D



(2017)

2.6 The radio telescope situated in Guizhou province has an effective aperture of 300 m for observations. It can be used to observe electromagnetic waves of frequencies between 7×10^7 Hz and 3×10^9 Hz. Estimate the minimum angular separation that the telescope can resolve.

A. 4.07×10^{-4} rad

B. 9.49×10^{-4} rad

C. 1.74×10^{-2} rad

D. 4.07×10^{-2} rad

A

B

C

D



(2017)

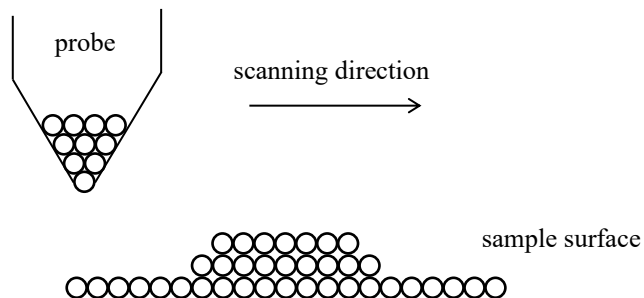
2.7 The leaves of the plant Edelweiss are covered with nanoscale filaments. These filaments absorb ultraviolet radiation but reflect all visible light. Which of the following statements are correct ?

- (1) The leaves appear white in colour under sunlight as the filaments reflect all visible light.
- (2) Optical microscopes cannot be used to observe the filaments.
- (3) Due to their tiny size, the filaments do not pose a health risk if absorbed into the human body.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

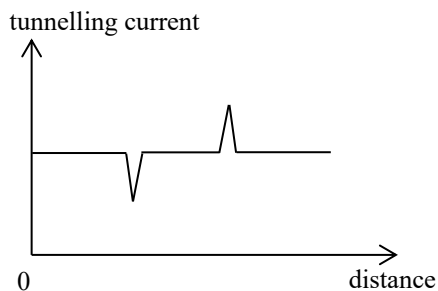
A	B	C	D
☐	☐	☐	☐

(2017) 2.8 A scanning tunnelling microscope (STM) scans across a sample surface as shown. The probe scans horizontally across the sample surface at a fixed height.

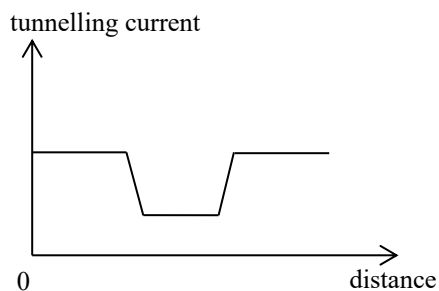


Which of the following graphs best represents the variation of the tunnelling current with distance travelled by the probe?

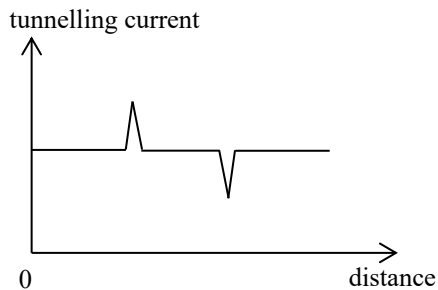
A.



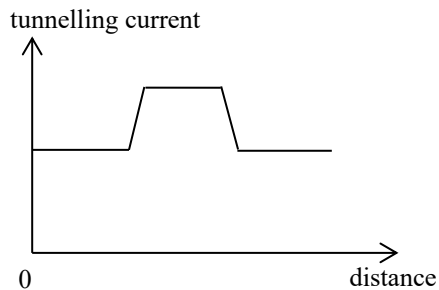
B.



C.

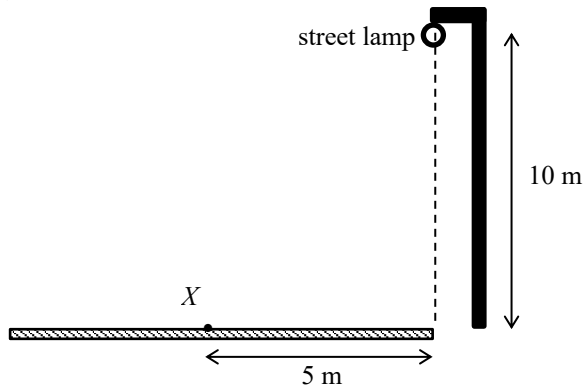


D.



A B C D
☐ ☐ ☐ ☐

- (2017) 3.1 A street lamp is installed 10 m above the ground on the side of a road. At a point X which is 5 m from the side on the road surface, the illuminance is 30 lux.



Taking the lamp as a point source emitting light uniformly in all directions, and neglecting the reflections and contributions from other light sources, estimate the luminous flux of the street lamp.

- A. 4.21×10^4 lm
B. 5.27×10^4 lm
C. 5.80×10^4 lm
D. 6.59×10^4 lm

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2017)

3.2 Arrange the efficacy of the following light sources from the largest to the smallest.

	<i>X</i>	<i>Y</i>	<i>Z</i>
rated power	11 W	13 W	20 W
luminous flux	300 lm	400 lm	500 lm

- A. *X, Y, Z*
- B. *X, Z, Y*
- C. *Y, X, Z*
- D. *Y, Z, X*

A B C D

☐ ☐ ☐ ☐

(2017)

3.3 The surface of an induction cooker is usually made of strengthened glass. The glass surface gets hot after use mainly because

- A. an induction cooker has a high energy efficiency.
- B. heat is transferred from the cooking utensil to the glass surface when the cooking utensil gets hot.
- C. an eddy current flows in the glass.
- D. heat is produced in the solenoid of the cooker.

A	B	C	D
☐	☐	☐	☐

(2017)

3.4 Which of the following changes **WILL NOT** reduce the Overall Thermal Transfer Value (OTTV) of a building ?

- A. building a roof-top garden
- B. installing heat insulating materials on the walls
- C. adding solar control window films to the windows
- D. painting the exterior of the building in dark colour

A

B

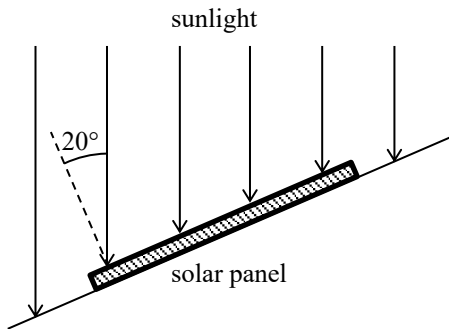
C

D



(2017)

- 3.5 A solar panel of area 3 m^2 is installed on a roof. Sunlight makes an angle of 20° to the normal of the panel at noon. The solar constant is 1366 W m^{-2} and 40% of the radiation power is absorbed by the atmosphere.



If the efficiency of the solar panel is 10%, what is the electrical power generated by it at noon ?

- A. 84 W
- B. 154 W
- C. 231 W
- D. 246 W

A	B	C	D
☐	☐	☐	☐

(2017)

3.6 The figure shows a wind turbine.



Which of the following statements explain why the wind turbine is **NOT** 100% efficient in converting the kinetic energy of the wind to electrical energy ?

- (1) There are mechanical energy losses in the moving parts.
- (2) Wind does not stop completely after passing through the rotor.
- (3) The direction of wind changes irregularly.

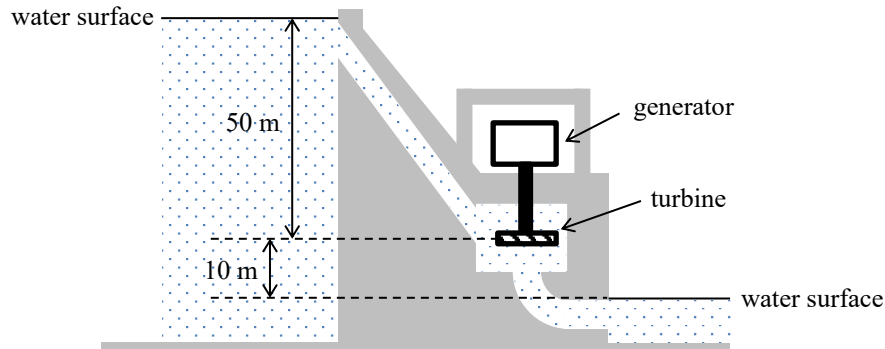
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2017)

- 3.7 The hydroelectric power plant shown has an efficiency of 40% in electricity generation. If the flow rate of the water is $300 \text{ m}^3 \text{ s}^{-1}$, what is the power output of the plant ?

Given: density of water is 1000 kg m^{-3} . Take $g = 9.81 \text{ m s}^{-2}$.

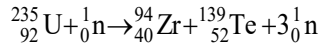


- A. 11.8 MW
B. 58.9 MW
C. 70.6 MW
D. 88.3 MW

- A B C D
☐ ☐ ☐ ☐

(2017)

3.8 Energy is released in the following nuclear fission of uranium-235.



Which of the following statements concerning the reaction is/are correct ?

- (1) The rate of the reaction can be controlled by absorbing some of the neutrons produced.
- (2) Mass is conserved in the reaction.
- (3) The binding energy per nucleon of ${}_{92}^{235}\text{U}$ is higher than that of ${}_{40}^{94}\text{Zr}$ or ${}_{52}^{139}\text{Te}$.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2018) 2.1 In Rutherford's scattering experiment, the distance of closest approach of an α particle to a gold atom can be deduced. This distance is a good estimate of the upper limit of

- A. the radius of an α particle.
- B. the radius of a gold atom.
- C. the radius of a gold nucleus.
- D. the thickness of the gold foil.



(2018)

2.2 Which of the following is/are the assumption(s) of the Bohr model of an atom ?

- (1) The orbital radii of the electrons are quantized.
- (2) The electric force between the nucleus and an electron provides the centripetal force for the circular motion.
- (3) The total energy of an orbiting electron remains unchanged.

A. (2) only

B. (3) only

C. (1) and (2) only

D. (1), (2) and (3)

A

B

C

D



(2018)

2.3 Which statements below about **spectra** are correct ?

- (1) A steel rod heated to white hot emits a continuous spectrum.
- (2) The dark lines in the absorption spectrum of an element match in frequency with the bright lines in this element's emission spectrum.
- (3) The atomic spectrum of an element is a piece of evidence for the existence of energy levels in its atoms.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2018)

2.4 According to the Bohr model of an atom when the electron of a hydrogen atom undergoes transition from an inner orbit to an orbit of larger radius, the hydrogen atom may have

- A. absorbed a photon, and the electron's kinetic energy decreases.
- B. absorbed a photon, and the electron's kinetic energy increases.
- C. emitted a photon, and the electron's kinetic energy decreases.
- D. emitted a photon, and the electron's kinetic energy increases.

A	B	C	D
☐	☐	☐	☐

(2018) 2.5 Some hydrogen atoms in the second excited state (i.e. $n = 3$) subsequently produces a spectrum consisting of a series of discrete lines. How many spectral lines of different wavelengths are there in the spectrum ?

- A. 1
- B. 2
- C. 3
- D. 4

A	B	C	D
☐	☐	☐	☐

(2018)

2.6. Electrons, each of mass m and charge e , are being accelerated in a transmission electron microscope (TEM). Estimate the accelerating voltage of the TEM if the de Broglie wavelength of the electrons is λ .

A. $\frac{h^2}{me\lambda^2}$

B. $\frac{h^2}{2me\lambda^2}$

C. $\frac{me\lambda^2}{h^2}$

D. $\frac{2me\lambda^2}{h^2}$

A

B

C

D



(2018)

2.7 Which of the following statements about a scanning tunnelling microscope (STM) is/are correct ?

- (1) It works like an optical microscope except that a high energy electron beam is used instead of visible light and a magnetic field acts as a lens.
- (2) It can be used to show the arrangement of atoms at a conductive surface.
- (3) The magnitude of the tunnelling current depends on the separation between the surface under investigation and the probe of the STM.

- A. (1) only
- B. (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2018)

2.8 Which statement below is **INCORRECT** ?

- A. Nano particles of silver show a colour different from the colour of silver in bulk form.
- B. Diamond is a poor thermal conductor and a poor electrical conductor as it does not have free electrons.
- C. A carbon nanotube is much stronger than steel of the same size.
- D. Carbon buckyballs such as C_{60} are formed by carbon atoms arranged in hollow cages.

A	B	C	D
☐	☐	☐	☐

(2018) 3.1 Hybrid cars are usually equipped with combustion engines with smaller power than those in petrol cars because

- (1) the combustion engine can be supplemented by the motor to provide the maximum power output of hybrid cars.
- (2) combustion engines in hybrid cars are more energy efficient than those in petrol cars.
- (3) the operation cost per unit power output of combustion engines with smaller power is lower.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

A

B

C

D



(2018)

3.2 Carbon dioxide emission associated with a passenger travelling in a MTR train for 1 km is about 50 g while that for a car is over 200 g. The possible reason(s) is/are

- (1) The MTR train and the car need energy to move the vehicle itself.
- (2) The energy efficiency in transporting a passenger for 1 km is higher in the case of the MTR train.
- (3) No burning of fossil fuel is involved in the production of energy for the MTR train.

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

A

B

C

D



(2018)

3.3 An air-conditioner with $\frac{\text{cooling capacity}}{\text{electrical power input}} = 2$ has a cooling capacity of 746 W. Estimate the rate at which thermal energy is being released to the environment outside.

- A. 373 W
- B. 746 W
- C. 1119 W
- D. 1492 W

A	B	C	D
☐	☐	☐	☐

(2018)

3.4 Which of the following contribute(s) to cooking with a microwave oven ?

- (1) The energy of microwave is only absorbed by the food's surface and then transferred to its interior by conduction.
- (2) Water molecules are polar due to non-uniform charge distribution within each molecule.
- (3) The electric field of microwave is continuously changing.

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

A

B

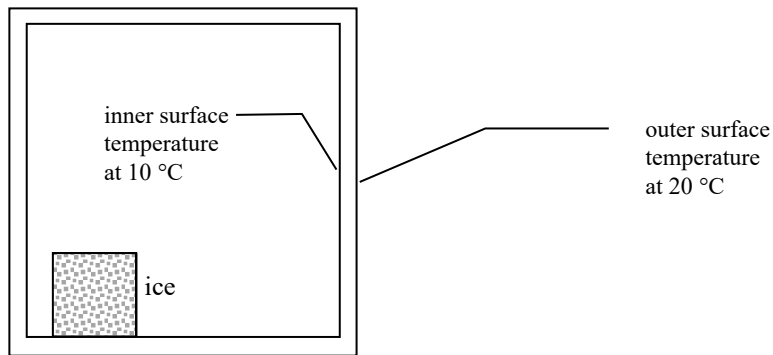
C

D



(2018)

- 3.5 A piece of ice is placed inside a closed plastic box. The wall of the plastic box is 2 cm thick and the thermal conductivity of plastic is $0.03 \text{ W m}^{-1} \text{ }^{\circ}\text{C}^{-1}$.



The temperature of the inner and the outer surfaces of the wall are 10°C and 20°C respectively. What is the rate of heat exchange (per unit area) between the inside and outside of the box and the direction of heat flow ?

	rate of heat exchange (per unit area)	direction of heat flow	A	B	C	D
A.	15 W m^{-2}	into the box	☐	☐	☐	☐
B.	15 W m^{-2}	out of the box	☐	☐	☐	☐
C.	36 W m^{-2}	into the box	☐	☐	☐	☐
D.	36 W m^{-2}	out of the box	☐	☐	☐	☐

(2018)

3.6 Which statement below about a house and its features **MUST BE** correct ?

- | | A | B | C | D |
|--|-----------------------|-----------------------|-----------------------|-----------------------|
| A. Thermal conduction is less efficient in materials that are darker in colour. | ☐ | ☐ | ☐ | ☐ |
| B. It is more desirable for the Overall Thermal Transfer Value (OTTV) of a house to be larger in cold areas. | | | | |
| C. Wall painted in white reduces the Overall Thermal Transfer Value (OTTV) of a house. | | | | |
| D. Wall painted in white increases the reflection of sunlight. | | | | |

(2018)

3.7 A wind turbine generator delivers a certain electric power with a wind blowing normal at speed v . If the length of the turbine blades is increased by 25% and the overall efficiency of the turbine generator remains the same, estimate the wind speed blowing normal to the turbine that gives the same electric power.

A. $0.59 v$

B. $0.64 v$

C. $0.86 v$

D. $0.93 v$

A

B

C

D



(2018)

3.8 In a nuclear reactor, uranium-235 is used as the fission fuel to generate electricity. When uranium-235 undergoes nuclear fission, 0.08% of its mass is converted to energy. If 20% of this energy becomes electrical energy, estimate the amount of uranium-235 used per second for generating 500 MW of electric power.

A. $1.4 \times 10^{-6} \text{ kg}$

B. $3.5 \times 10^{-5} \text{ kg}$

C. $8.2 \times 10^{-3} \text{ kg}$

D. $1.0 \times 10^{-2} \text{ kg}$

A

B

C

D



(2019) 2.1 There are dark lines in the Sun's spectrum because lights at certain wavelengths emitted by the Sun are .

A. completely absorbed by the Sun's atmosphere.

B. completely absorbed by the Earth's atmosphere.

C. partly absorbed by the Sun's atmosphere.

D. partly absorbed by the Earth's atmosphere.

A

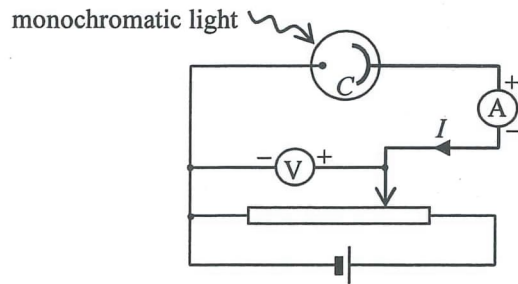
B

C

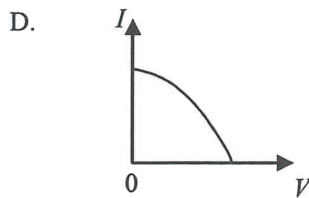
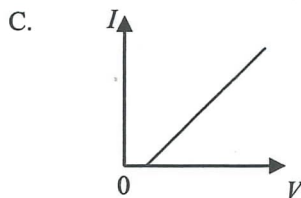
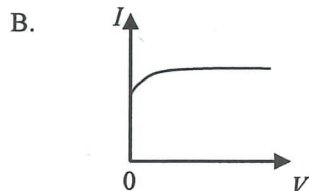
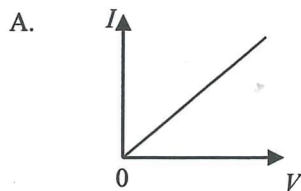
D



(2019) 2.2 The set-up below is for investigating the maximum kinetic energy of photoelectrons in photoelectric effect.



Monochromatic light of fixed intensity is shone on the cathode C of a photocell. The p.d. V applied across the photocell is adjusted and the corresponding current I is measured. What will be the graph of I against V ?



A
☐

B
☐

C
☐

D
☐

(2019) 2.3 What is the maximum kinetic energy of the photoelectrons emitted if gold is illuminated by electromagnetic radiation of wavelength 200 nm ? The work function of gold is 5.30 eV.

A. 0.916 eV

B. 5.30 eV

C. 6.22 eV

D. 11.3 eV

A

B

C

D



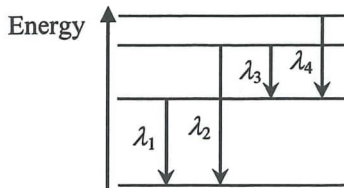
(2019) 2.4 Which microscope(s) below can be used to manipulate individual atoms on conductive surfaces ?

- (1) optical microscope
- (2) scanning tunneling microscope
- (3) transmission electron microscope

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2019) 2.5 The energy diagram for an atom is shown below.



The electron transitions shown give rise to emission lines of wavelengths λ_1 , λ_2 , λ_3 and λ_4 respectively. Which of the following is/are correct ?

(1) $\frac{1}{\lambda_3} < \frac{1}{\lambda_4}$

(2) $\lambda_1 < \lambda_2$

(3) $\lambda_1 + \lambda_3 = \lambda_2$

A. (1) only

B. (2) only

C. (1) and (2) only

D. (2) and (3) only

A

B

C

D



(2019) 2.6 The de Broglie wavelength of a neutron with kinetic energy 1 keV is λ . If the kinetic energy of the neutron becomes 2 keV, what is its de Broglie wavelength ?

A. $\sqrt{2} \lambda$

B. λ

C. $\frac{\lambda}{2}$

D. $\frac{\lambda}{\sqrt{2}}$

A

B

C

D



(2019) 2.7 Two point sources of red light at a distance of 160 m from an observer can just be resolved by the naked eyes. If they are replaced by point sources of violet light, how should the observer move from the original position such that the two sources can just be resolved ?

- A. move about 280 m further away from the sources
- B. move about 120 m further away from the sources
- C. move about 120 m towards the sources
- D. move about 70 m towards the sources

A	B	C	D
☐	☐	☐	☐

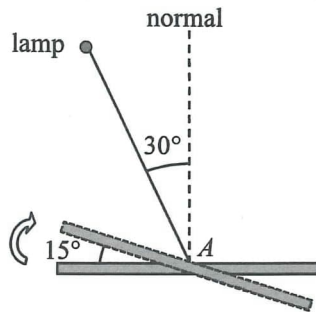
(2019) 2.8 A metal is broken up into nano-sized particles which are then closely packed. Which of the following statements is/are correct ?

- (1) The total volume remains more or less unchanged but the total surface area increases.
- (2) The shape changes but the arrangement of atoms remains unchanged.
- (3) The chemical properties change but the physical properties remain unchanged.

- A. (1) only
- B. (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

- (2019) 3.1 A book in a horizontal position is underneath a lamp positioned at an angle of 30° with the normal to the book as shown. The illuminance on the book around point A is 10 lx. Estimate the illuminance around A when the book is tilted about A by 15° .



- A. 5.44 lx
B. 8.16 lx
C. 9.66 lx
D. 12.2 lx

A B C D
☐ ☐ ☐ ☐

(2019) 3.2 The battery pack of an electric vehicle can store 25 kW h of energy when fully charged. The output mechanical power of the vehicle is 12.5 kW and its end-use energy efficiency is 80%. Estimate the maximum duration of travel of the vehicle at this output power.

- A. 1.3 hours
- B. 1.6 hours
- C. 2 hours
- D. 2.5 hours

A	B	C	D
☐	☐	☐	☐

(2019) 3.3 Which of the following descriptions about a hybrid car is/are correct ?

- (1) The motor and the combustion engine of a hybrid car can be turned on at the same time to drive the car.
- (2) A hybrid car is said to be environmental friendly as it does not emit pollutants directly.
- (3) If the battery of a hybrid car cannot be charged via a wall socket, it can only be charged through the regenerative braking system during deceleration.

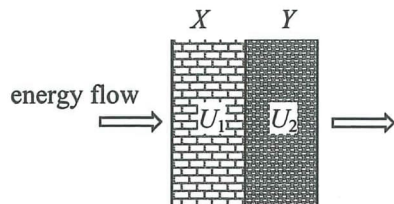
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

- (2019) 3.4 Air-conditioners P and Q below are used respectively for cooling two identical rooms under the same environmental conditions. Which statement is **INCORRECT** ?

air-conditioner	cooling capacity / kW	coefficient of performance (COP)
P	3.2	2.2
Q	3.0	1.9

- | | | | | | |
|----|---|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | The room with air-conditioner P cools down quicker. | A | B | C | D |
| B. | P is more energy efficient. | ☐ | ☐ | ☐ | ☐ |
| C. | The rate of releasing total amount of thermal energy to the outside is larger for P . | | | | |
| D. | P consumes more power. | | | | |



A wall is composed of layers X and Y of U-values U_1 and U_2 respectively. Both layers have the same thickness and dimensions, and there is no air gap between them. Which expression gives the U-value of the wall ?

A. $U_1 + U_2$

B. $\frac{1}{2}(U_1 + U_2)$

C. $\frac{2U_1U_2}{U_1 + U_2}$

D. $\frac{U_1U_2}{U_1 + U_2}$

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2019) 3.6 Which of the following arrangements can reduce the Overall Thermal Transfer Value (OTTV) of a building ?

- (1) adding heat insulating materials onto the building envelope
- (2) introducing air space in walls
- (3) replacing concrete walls with glass windows

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2019) 3.7 A wind turbine delivers a power of 800 W at a wind speed of 4 m s^{-1} blowing normal to its blades. Estimate the power delivered by the turbine if the wind speed increases to 6 m s^{-1} blowing in the same direction.

A. 1200 W

B. 1800 W

C. 2700 W

D. 3200 W

A



B

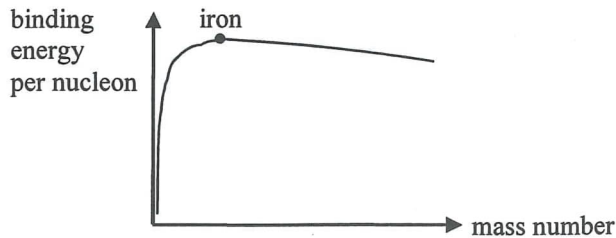


C



D





Which of the following can be directly inferred from the above binding energy curve ?

- (1) The negative slope of the curve indicates that the total mass of the nuclides produced after fission is larger than the mass of the heavy nucleus before fission.
- (2) The positive steep slope of the curve characterises that nuclear fusion in general produces much more energy per nucleon than nuclear fission.
- (3) Iron being at the peak of the curve indicates that this element has the most stable atomic nucleus.

- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

A

B

C

D



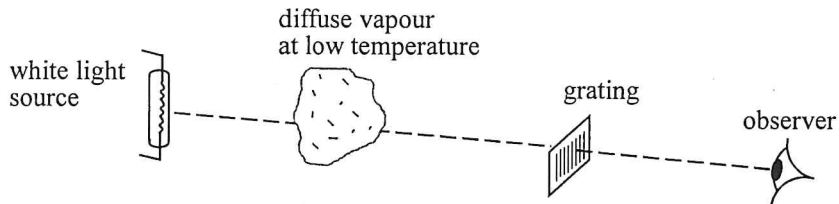
(2020)

2.1 Which of the following statements about Rutherford's scattering experiment is/are correct ?

- (1) Thin gold foil was used so that each α particle would most likely be scattered by one gold nucleus only.
- (2) If β particles were directed towards the gold foil instead of α particles, they could be deflected by both the gold nuclei and the electrons in the foil.
- (3) Gold foil was used because gold contains free electrons.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐



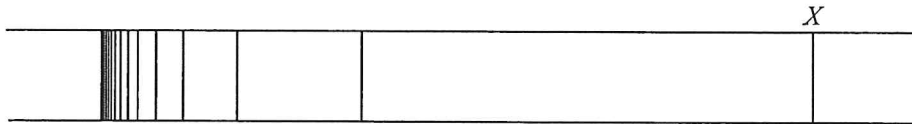
In the experiment shown above, dark lines are observed on a colour background. Which explanation is correct ?

- A. No photons corresponding to the dark lines were emitted from the source.
- B. No photons corresponding to the dark lines were emitted from the vapour.
- C. Photons corresponding to the dark lines from the source were absorbed by the vapour.
- D. Photons corresponding to the dark lines from the source were scattered back to the source by the vapour.

A	B	C	D
☐	☐	☐	☐

(2020)

- 2.3 The figure below shows the emission lines of the hydrogen spectrum associated with electron transitions from excited energy levels to the ground state ($n = 1$). It is known that the energy level of the ground state is $E_1 = -13.6$ eV.



Which statement is correct ?

- A. Spectral line X has the highest frequency.
- B. The shortest wavelength of the lines in this spectral series is about 90 nm.
- C. These spectral lines are in the infra-red region.
- D. Some of these lines also appear in other spectral series of the hydrogen spectrum.

A	B	C	D
☐	☐	☐	☐

(2020) 2.4 Which statements about the Bohr atom model are correct ?

- (1) It can explain why α particles can be rebounded by a thin gold foil.
- (2) It can give the atomic spectra of a singly ionized helium atom (He^+).
- (3) A postulate of this model is that the angular momentum of the electron in an hydrogen atom is quantized.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2020)

2.5 An atom has three energy levels : X , Y , and Z . When the atom transits from X to Y , it emits a photon of wavelength λ_1 . The atom transits from Y to Z when it absorbs a photon of wavelength λ_2 , with $\lambda_1 > \lambda_2$. These energy levels arranged in descending order is

- A. $Z\ X\ Y$.
- B. $Z\ Y\ X$.
- C. $X\ Z\ Y$.
- D. $X\ Y\ Z$.



(2020) 2.6 An electron beam accelerated through a potential difference of V is directed to a thin crystal layer. The diffraction pattern obtained is similar to that produced by X-rays of wavelength λ . What potential difference should be used for accelerating the electron beam so as to give a diffraction pattern similar to that produced by X-rays of wavelength 2λ ?

- A. $\frac{V}{4}$
B. $\frac{V}{2}$
C. $2V$
D. $4V$

A	B	C	D
☐	☐	☐	☐

(2020)

2.7 Which of the following statements about a scanning tunnelling microscope (STM) is/are correct ?

- (1) STM can reveal the internal structure of a specimen as three-dimensional images are produced.
- (2) In STM imaging, the surface of the specimen must be electrically conducting.
- (3) The resolving power of STM is limited by Rayleigh criterion.

- A. (1) only
- B. (2) only
- C. (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

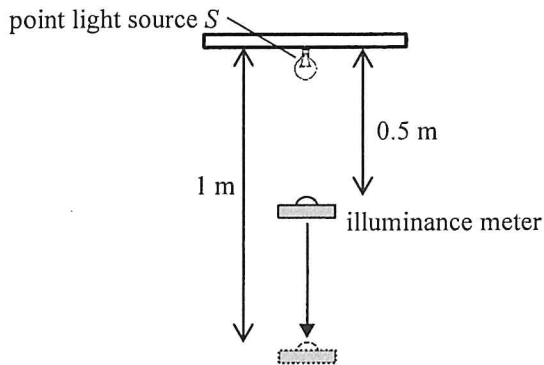
(2020) 2.8 Which of the following statements about materials in **bulk form** and in **nano size** are correct ?

- (1) Most materials exhibit different colour in the states mentioned above.
- (2) Most materials in nano size have lower melting point.
- (3) Materials in nano size are usually more efficient in serving as catalyst.

A.	(1) and (2) only	A	B	C	D
B.	(1) and (3) only	☐	☐	☐	☐
C.	(2) and (3) only				
D.	(1), (2) and (3)				

(2020)

- 3.1 An illuminance meter is placed 0.5 m directly underneath a point light source S emitting a certain luminous flux as shown.



If the luminous flux emitted by S is doubled and the meter is lowered to a position 1 m underneath S , what would be the change in the meter reading ?

- A. decreases by 25%
- B. decreases by 50%
- C. remains unchanged
- D. increases by 50%

A

B

C

D



(2020)

- 3.2 A solar furnace uses a reflector of area 9 m^2 to collect sunlight so as to heat up a piece of iron of mass 2 kg . The efficiency of the furnace is 50% . Estimate the time taken to heat up the piece of iron from 30°C to 90°C .
Given: specific heat capacity of iron = $450 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$
solar power per unit area received on the Earth's surface = 1000 W m^{-2}

- A. 12 s
B. 24 s
C. 48 s
D. 96 s

A	B	C	D
☐	☐	☐	☐

(2020)

3.3 An electric vehicle's battery of capacity 40 kW h is completely discharged initially. It is charged with a terminal voltage of 220 V at an average current of 32 A. Estimate the time required to fully charge this battery. Assume that 20% energy loss occurs during charging.

A. 4.6 hours

B. 5.7 hours

C. 6.8 hours

D. 7.1 hours

A



B



C



D



(2020) 3.4 What are the reasons for a microwave oven having an 'end-use energy efficiency' of less than 100% ?

- (1) Some energy is lost during the transmission of electrical energy from the power plant to the oven.
- (2) The microwave oven cannot convert all the electrical energy to the energy of microwave.
- (3) Some microwaves are absorbed by the oven's body and does not reach the food.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2020) 3.5 During summer time, heat flows into a house through its walls of a certain thickness. The rate of heat flow per unit area of the wall would be reduced if

- (1) the temperature difference between the outer and inner surfaces of the wall is smaller.
- (2) the thickness of the wall is increased.
- (3) material of a larger U-value is used for building the wall.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

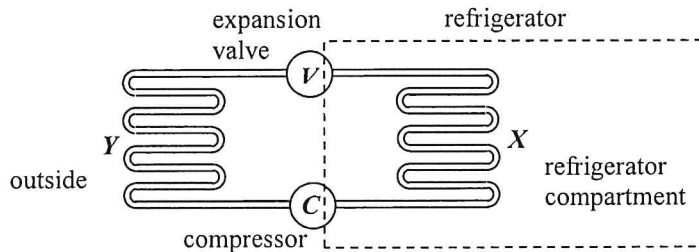
(2020) 3.6 A room is kept cool by an air-conditioner of cooling capacity P . The temperatures inside and outside the room are $27\text{ }^{\circ}\text{C}$ and $31\text{ }^{\circ}\text{C}$ respectively. The rate of heat flowing into the room by radiation through windows and that by conduction are in the ratio 1:4. If the cooling capacity is raised to $2P$ while the temperature outside the room is still $31\text{ }^{\circ}\text{C}$, estimate the temperature inside the room. Assume that the rate of heat flowing into the room by radiation is unchanged.

- A. $21\text{ }^{\circ}\text{C}$
- B. $22\text{ }^{\circ}\text{C}$
- C. $23\text{ }^{\circ}\text{C}$
- D. $25\text{ }^{\circ}\text{C}$

A	B	C	D
☐	☐	☐	☐

(2020)

3.7 A simplified schematic diagram of a refrigerator is as shown.



What is the direction of flow of the refrigerant through the expansion valve V ? Which component, X or Y , contains refrigerant at a higher temperature?

	direction of flow of the refrigerant through the expansion valve	component that contains refrigerant at a higher temperature	A	B	C	D
A.	$X \rightarrow V \rightarrow Y$	X				
B.	$X \rightarrow V \rightarrow Y$	Y	☐	☐	☐	☐
C.	$Y \rightarrow V \rightarrow X$	X				
D.	$Y \rightarrow V \rightarrow X$	Y				

(2020)

3.8 Which of the following is/are 'renewable energy source(s)' ?

- (1) wind power
- (2) natural gas
- (3) nuclear power

- A. (1) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A

B

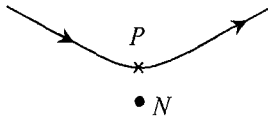
C

D



(2021)

- 2.1 The path of an α particle approaching a massive nucleus at N is shown below. At point P the α particle is closest to the nucleus.



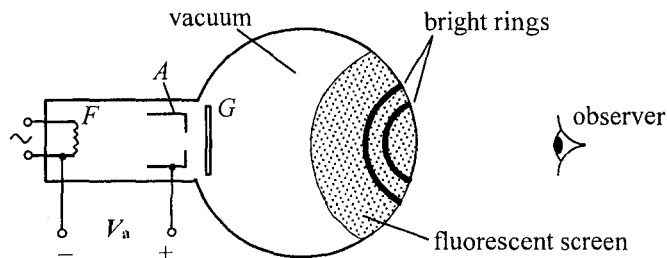
Which statement below is correct ?

- A. At P the kinetic energy of the α particle is at a maximum.
- B. At P the total energy of the α particle is at a minimum.
- C. If the nucleus has a larger atomic number, the distance between P and N would be larger.
- D. If the α particle has a larger initial kinetic energy, the distance between P and N would be larger.

A	B	C	D
☐	☐	☐	☐

(2021)

2.2 The figure shows an electron diffraction tube that can reveal the nature of electrons.



Electrons liberated from a heated filament F are accelerated by a high voltage V_a between F and anode A . The electrons then pass through a thin graphite film G and form bright and dark concentric rings on a fluorescent screen as shown. Which descriptions about this experiment are correct ?

- (1) It demonstrates the wave nature of fast-moving electrons.
- (2) Electrons are diffracted by the graphite film.
- (3) If V_a increases slightly, the radii of the rings would increase.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2021)

- 2.3 When monochromatic lights of wavelengths λ and 2λ are incident on a metal surface, the maximum kinetic energies of the photoelectrons emitted are in the ratio of 3:1. Find the longest wavelength of monochromatic light that can trigger photoemission for such metal.

A. $\frac{5\lambda}{2}$

B. 3λ

C. $\frac{7\lambda}{2}$

D. 4λ

A



B



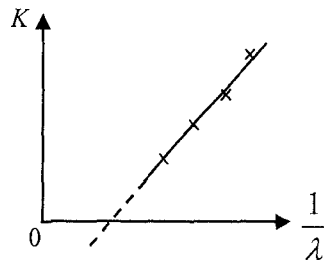
C



D



- (2021) 2.4 The graph shows the variation of the maximum kinetic energy K of the photoelectrons emitted from a certain metal with the reciprocal of the wavelength $1/\lambda$ of the incident light.

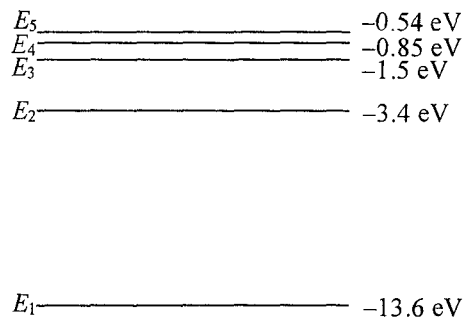


How would the graph change if incident light of lower intensity is shone on another metal having a smaller work function?

	slope of the graph	intercept on horizontal axis	A	B	C	D
A.	unchanged	larger	☐	☐	☐	☐
B.	unchanged	smaller	☐	☐	☐	☐
C.	smaller	larger	☐	☐	☐	☐
D.	larger	smaller	☐	☐	☐	☐

(2021)

2.5



The figure shows the five lowest energy levels of a hydrogen atom. If electron transition from E_4 to E_2 emits a photon of blue light, which electron transition below would emit red light ?

Given: the visible spectrum is about 400 nm to 750 nm

- A. E_5 to E_2
- B. E_4 to E_3
- C. E_3 to E_2
- D. E_2 to E_1

A	B	C	D
☐	☐	☐	☐

(2021)

2.6 When an electron of mass m and charge e is accelerated from rest by a voltage V , its de Broglie wavelength λ is given by $\lambda = \frac{h}{\sqrt{2meV}}$, where h is the Planck constant. If λ is expressed in nanometre (nm) and V in kilovolt (kV), then λ is approximately equal to

A. $\frac{0.04}{\sqrt{V}}$.

B. $\frac{0.12}{\sqrt{V}}$.

C. $\frac{0.4}{\sqrt{V}}$.

D. $\frac{1.2}{\sqrt{V}}$.

A	B	C	D
☐	☐	☐	☐

(2021)

2.7 Which statements about optical microscope and transmission electron microscope (TEM) are correct ?

- (1) The higher resolving power of TEM is enabled by the much shorter wavelength of its electron beam than that of visible light employed in an optical microscope.
- (2) The current-carrying coils in a TEM provide magnetic fields for converging the electron beam, which is similar to the lenses in an optical microscope for converging light.
- (3) The angular resolution of both microscopes are limited by the Rayleigh's criterion.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A

B

C

D



(2021)

2.8 A nano material

- (1) has a higher volume to surface area ratio than the same substance in bulk form.
- (2) has at least one dimension less than 1 nm.
- (3) is chemically more reactive than the same substance in bulk form.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

A

B

C

D



(2021)

- 3.1 A light power of 1 W delivered by a green light source corresponds to a luminous flux of 683 lm. Taking into account the sensitivity of human eye, a light power of 1 W delivered by a filament lamp emitting white light only gives about half of this luminous flux. If the end-use energy efficiency of the filament lamp is about 3%, estimate its efficacy.

- A. 40 lm W⁻¹
- B. 20 lm W⁻¹
- C. 10 lm W⁻¹
- D. 5 lm W⁻¹

A	B	C	D
☐	☐	☐	☐

(2021)

3.2 A wind turbine generator can extract energy from moving air. However, the kinetic energy of wind cannot be fully transformed into electrical energy because

- (1) wind velocity cannot be zero after passing through the turbine.
- (2) there is loss in transformation of energy in the generator.
- (3) wind may not always blow in the direction normal to the turbine.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2021)

3.3 Even on a clear day, the atmosphere absorbs at least 26.8% of solar power. Find the maximum power output of a solar panel of area 5 m^2 which has an efficiency of 15%. Given: solar constant = 1366 W m^{-2}

- A. 275 W
- B. 750 W
- C. 1560 W
- D. 4250 W

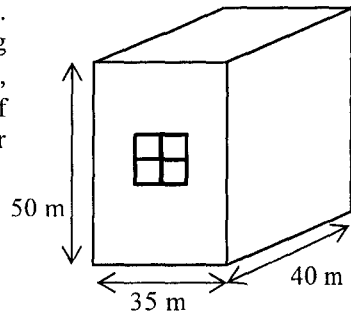
A	B	C	D
☐	☐	☐	☐

(2021)

- 3.4 The figure shows a concrete building of dimensions $35\text{ m} \times 40\text{ m} \times 50\text{ m}$. It is given that the Overall Thermal Transfer Value (OTTV) of a building should not exceed 24 W m^{-2} . Find the maximum number of windows, each of size $2\text{ m} \times 3\text{ m}$, that can be installed on the walls of the building if the equivalent temperature difference between the interior and the exterior of the building is 10°C .

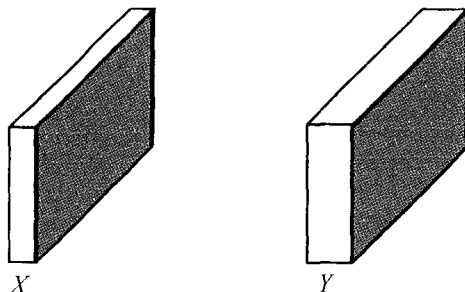
Given: U-value of the concrete of the building = $2.0\text{ W m}^{-2}\text{ K}^{-1}$

U-value of the glass of the windows = $5.7\text{ W m}^{-2}\text{ K}^{-1}$



- A. 960
- B. 598
- C. 160
- D. 120

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |



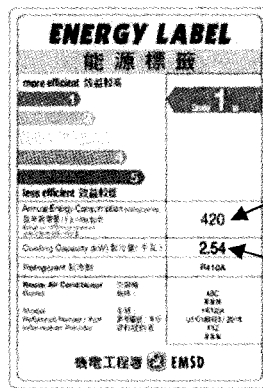
Walls X and Y having the same area are made of the same material, with Y thicker than X . If the temperature difference between the two faces of each wall is the same, X and Y have the same

- (1) thermal conductivity.
- (2) thermal transmittance (U-value).
- (3) rate of heat transfer by conduction.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2021) 3.6 The energy label below indicates the information of a certain air-conditioner.



energy consumption of 420 kW h
for 1200 hours operation per year

cooling capacity
2.54 kW

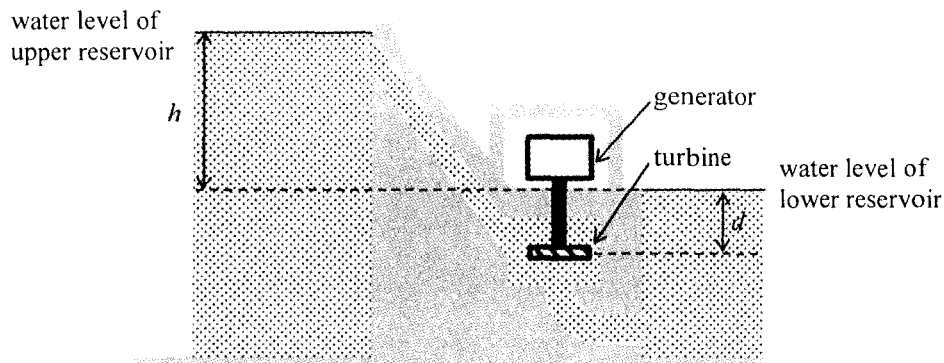
Find the COP (coefficient of performance) of this air-conditioner.

- A. 1.12
- B. 1.38
- C. 7.26
- D. 8.89

- A ☐
- B ☐
- C ☐
- D ☐

(2021)

3.7 The figure below shows a hydroelectric power plant.



Which factors below can affect the maximum power output of the plant ?

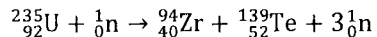
- (1) The height difference between the water levels in the upper and lower reservoirs, h .
- (2) The distance between the turbine and the water level of the lower reservoir, d .
- (3) The rate of water flowing through the turbine.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2021)

3.8 For the fission reaction of a U-235 nucleus shown below, the mass defect is 0.1855u.



How much energy (in J) would be produced when 1 kg of U-235 completely undergoes such fission ?

Given: molar mass of U-235 = 235 g

1.49×10^{-10} J of energy is released for a mass defect of 1 u

A. $\frac{1000}{235} \times 6.02 \times 10^{23} \times 0.1855 \times 1.49 \times 10^{-10}$

A

☐

B

☐

C

☐

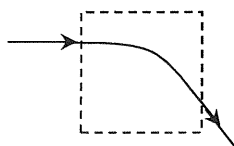
D

☐

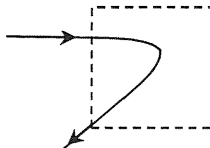
B. $\frac{1}{235} \times 6.02 \times 10^{23} \times 0.1855 \times 1.49 \times 10^{-10}$

C. $\frac{235}{1000} \times 6.02 \times 10^{23} \times 0.1855 \times 1.49 \times 10^{-10}$

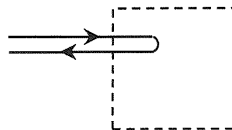
D. $\frac{1000}{235} \times 6.02 \times 10^{23} \times 1.49 \times 10^{-10}$



(1)



(2)



(3)

Each of the above figures shows the path of an α particle scattered by a fixed charged particle P (**NOT** shown in the figure but inside the dotted region). If electrostatic force is the only interaction between the two particles, from which figure can one deduce that P **must be** positively charged?

- A. (3) only
 B. (1) and (2) only
 C. (2) and (3) only
 D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

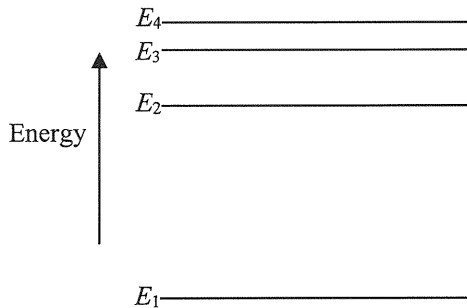
(2022) 2.2 Which statements below about the **absorption spectrum** of helium are correct ?

- (1) Helium atoms can only absorb photons with energy that equals the difference between two energy levels of helium.
- (2) The dark lines in the absorption spectrum of helium match those in its emission spectrum.
- (3) The dark lines in the spectrum correspond to the wavelength of photons being absorbed by helium atoms.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

- (2022) 2.3 The diagram represents the energy levels **drawn to scale** for an electron in a certain atom. The electron transition from E_3 to E_1 produces a green line. What electron transition would give rise to a red line ?
Given: the visible spectrum is about 400 nm to 750 nm



- A. E_3 to E_2
B. E_4 to E_2
C. E_2 to E_1
D. E_4 to E_1

A



B



C



D



(2022) 2.4 According to the Bohr model of the hydrogen atom, which of the following physical quantities about the electron of the atom will decrease when a photon is emitted from the hydrogen atom ?

- (1) the orbital radius of the electron
- (2) the angular momentum of the electron
- (3) the kinetic energy of the electron

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A

B

C

D



- (2022) 2.5 An optical microscope is employed to view a specimen such that two points of separation 3×10^{-6} m can just be resolved. If the diameter of the objective lens is 10 mm, deduce a probable combination of objective-to-specimen separation L and wavelength λ of the light used.

	L / cm	λ / nm				
A.	5	410	A	B	C	D
B.	4	615	☐	☐	☐	☐
C.	3	410				
D.	2	615				

(2022) 2.6 A voltage V_0 is applied to the anode of a transmission electron microscope (TEM) in order to produce an electron beam of de Broglie wavelength 0.01 nm. What should the anode voltage be for reducing the wavelength to 0.005 nm ?

A. $\frac{V_0}{4}$

B. $\frac{V_0}{2}$

C. $2V_0$

D. $4V_0$

A



B



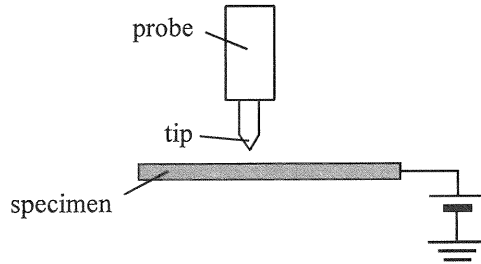
C



D



(2022) 2.7 The simplified schematic diagram below shows a scanning tunnelling microscope (STM).



Which of the following statements about its operation is/are correct ?

- (1) The tunnelling current increases with the conductivity of the specimen's surface.
- (2) The tunnelling current is independent of the medium between the tip and the specimen.
- (3) When scanning in the constant height mode, the change in distance between the tip and the specimen's surface is in atomic scale.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2022)

2.8 Which of the following is/are in nanoscale ?

- (1) the diameter of a human hair
- (2) the size of a gold nucleus
- (3) the size of a coronavirus (e.g. COVID-19) particle

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

A

B

C

D



(2022) 3.1 Three types of lamps, P , Q and R , are available.

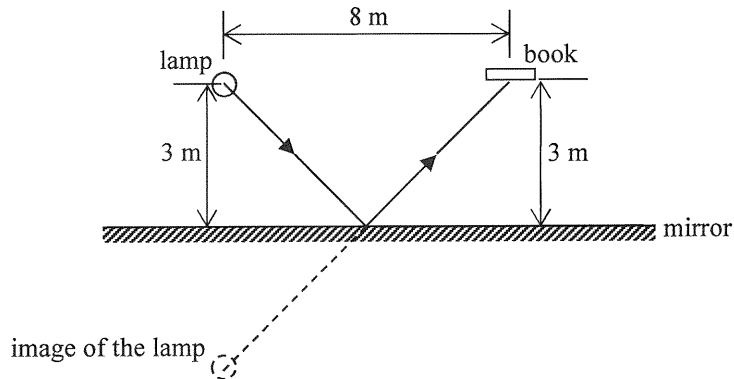
lamp	power rating / W	efficacy / lm W^{-1}
P	6	100
Q	8	110
R	10	85

When they operate at their respective ratings under the same conditions, which one is the brightest ? Which one consumes the least amount of energy ?

- | | | | | | | |
|----|-----------|---------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| | brightest | least amount of energy consumed | | | | |
| A. | Q | P | A | B | C | D |
| B. | Q | Q | ☐ | ☐ | ☐ | ☐ |
| C. | R | P | | | | |
| D. | R | Q | | | | |

(2022)

- 3.2 The figure shows a lamp of luminous flux 6000 lm placed 3 m above a horizontal mirror, and a book is placed horizontally at the same height as the lamp. The light from the lamp is reflected from the mirror to the book. If the mirror reflects 100% of the light energy, find the illuminance on the side of the book facing the mirror.



- A. 2.9 lx
B. 3.8 lx
C. 4.8 lx
D. 11.5 lx

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2022)

- 3.3 The upper and lower reservoirs of a hydroelectric power plant have a height difference of 80 m in water levels. The turbine of the power plant gives a total output power of 1000 MW. Assuming that the overall efficiency of the hydroelectric power plant is 80%, estimate the water flow rate, in kg s^{-1} , through the turbine.
($g = 9.81 \text{ m s}^{-2}$)

- A. 1.02×10^6
B. 1.27×10^6
C. 1.52×10^6
D. 1.59×10^6

A	B	C	D
☐	☐	☐	☐

(2022) 3.4 According to the Energy Efficiency Labelling Scheme (EELS), which statement below about an air-conditioner of grade 1 is correct ?

- | | A | B | C | D |
|--|-----------------------|-----------------------|-----------------------|-----------------------|
| A. It is more energy efficient than a refrigerator of grade 5. | | | | |
| B. Its electric power consumption must be less than that of an air-conditioner of grade 5. | ☐ | ☐ | ☐ | ☐ |
| C. Its energy efficiency must be above the average energy efficiency of air-conditioners in the same category of cooling capacity. | | | | |
| D. Its energy efficiency must be below the average energy efficiency of air-conditioners in the same category of cooling capacity. | | | | |

(2022) 3.5 The COP (coefficient of performance) of an air-conditioner is 4.0. If it is replaced by another one with the same input power and of COP 6.0, the percentage increase in the **rate of total thermal energy released** to the environment outside is

A. 29 % .

B. 33 % .

C. 40 % .

D. 50 % .

A

B

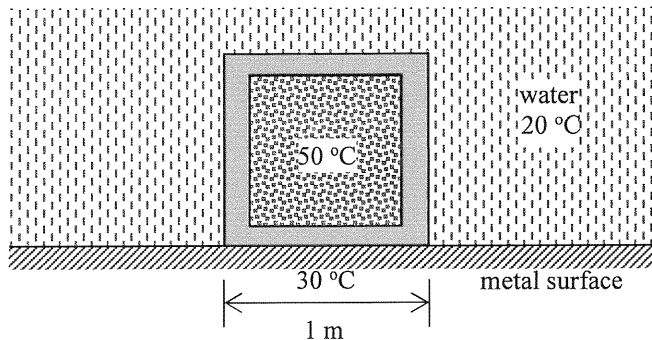
C

D



(2022)

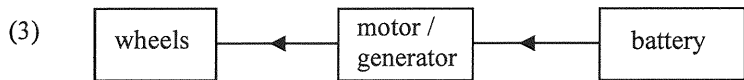
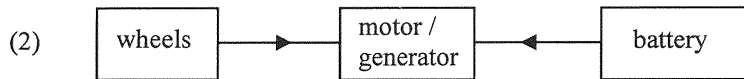
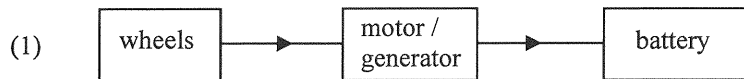
- 3.6 The figure shows a cubical container of side length 1 m submerged in water of temperature $20\text{ }^{\circ}\text{C}$ and its base rested on a metal surface at $30\text{ }^{\circ}\text{C}$. The container is filled with a certain liquid at $50\text{ }^{\circ}\text{C}$. The U-value of the walls of the container is $10\text{ W m}^{-2}\text{ K}^{-1}$. Find the rate of thermal energy transferred away from the container at the instant shown.



- A. 900 W
B. 1100 W
C. 1300 W
D. 1700 W

- | A | B | C | D |
|-----------------------|-----------------------|-----------------------|-----------------------|
| ☐ | ☐ | ☐ | ☐ |

(2022) 3.7 In each of the schematic diagrams below, the arrows represent the directions of power flow in an electric vehicle's power system.



Which diagram most probably corresponds to the situation when the vehicle is (i) accelerating and (ii) braking ?

	accelerating	braking				
A.	(3)	(2)	A	B	C	D
B.	(3)	(1)	○	○	○	○
C.	(1)	(2)				
D.	(1)	(3)				

(2022) 3.8 What are the factors that determine the power generated by a wind turbine ?

- (1) air density
- (2) wind speed
- (3) length of the turbine blades

- | | | | | | |
|----|------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | (1) and (2) only | A | B | C | D |
| B. | (1) and (3) only | ☐ | ☐ | ☐ | ☐ |
| C. | (2) and (3) only | | | | |
| D. | (1), (2) and (3) | | | | |

(2023) 2.1 In a Rutherford scattering experiment, most α particles can pass straight through the gold foil while a few being deflected significantly. This proves that an atom has a tiny massive nucleus. Suppose the mass of an atom is evenly distributed within the whole atom,

- A. most α particles would be stopped by the gold foil.
- B. nearly all α particles would pass straight through the gold foil.
- C. most α particles would be deflected significantly and only a few would be able to pass straight through the gold foil.
- D. most α particles would rebound by 180° .

A	B	C	D
☐	☐	☐	☐

(2023) 2.2 Which statements about atomic spectra are correct ?

- (1) Atomic spectra are due to discrete energy levels for electrons in atoms.
- (2) Absorption lines in atomic spectra arise when orbital electrons gain energy.
- (3) Emission lines in atomic spectra only correspond to the visible part of the electromagnetic spectrum.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2023) 2.3 A hydrogen atom at ground state has an ionization energy of 13.6 eV. Which of the following photons can be absorbed by this hydrogen atom ?

- (1) 3.4 eV
- (2) 10.2 eV
- (3) 13.7 eV

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2023) 2.4 A monochromatic light is used to illuminate the cathodes of two photocells X and Y , with each photocell connected to a sensitive galvanometer. The galvanometer connected to X shows a reading while that connected to Y does not. Which statements are correct ?

- (1) Electrons can escape from the cathode of X but not from the cathode of Y .
- (2) Compared to Y , the metallic cathode of X has a smaller threshold frequency.
- (3) Compared to Y , the metallic cathode of X has a larger work function.

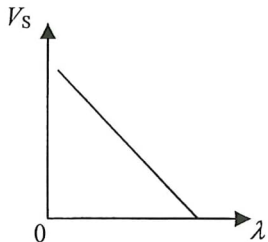
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

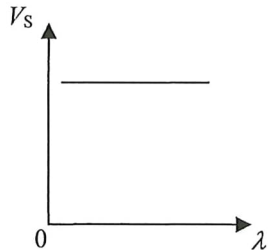
(2023) 2.5

In a photoelectric experiment, monochromatic lights of different wavelength λ are incident upon the cathode of the same photocell and the corresponding stopping potential V_s for the photoelectrons emitted is measured. What will be the graph of V_s against λ ?

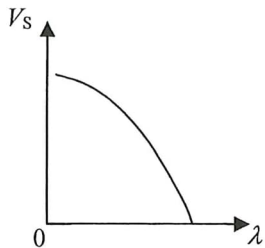
A.



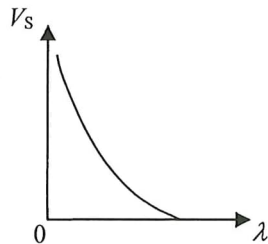
B.



C.



D.



A
☐

B
☐

C
☐

D
☐

(2023)

2.6 A beam of fast-moving electrons shows wave properties and its de Broglie wavelength is 0.10 nm. What is the average kinetic energy of the beam of electrons ?

A. 151 eV

B. 302 eV

C. 6.2 keV

D. 12.4 keV

A

B

C

D



(2023)

2.7

The angular resolution limit of a transmission electron microscope (TEM) when operated with anode voltage 5 kV is about 4×10^{-10} rad. What is the angular resolution limit of this TEM if it is operated with anode voltage 20 kV ? Assume that its resolution is limited by diffraction only.

A. 1×10^{-10} rad

B. 2×10^{-10} rad

C. 8×10^{-10} rad

D. 16×10^{-10} rad

A

B

C

D

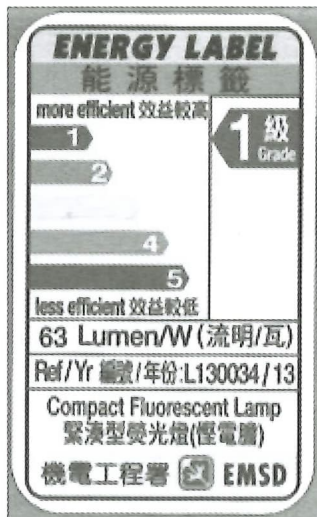


(2023) 2.8 Which phenomena below is/are due to the fact that materials in nanoscale have a much greater surface effect than in their bulk form ?

- (1) Titanium dioxide (TiO_2) in nanoscale is a more efficient catalyst.
- (2) Silver in nano-sized form has a lower melting point.
- (3) An electric current can flow between two surfaces separated at nanoscale even though they are not in contact.

- | | | | | | |
|----|------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | (1) only | A | B | C | D |
| B. | (3) only | ☐ | ☐ | ☐ | ☐ |
| C. | (1) and (2) only | | | | |
| D. | (2) and (3) only | | | | |

(2023) 3.1 The energy label of a 12 W compact fluorescent lamp is shown below.



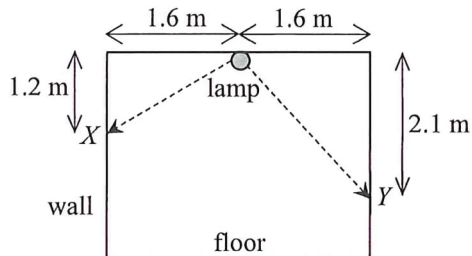
Which of the following statements are correct ?

- (1) The efficacy of the lamp is 63 lm W^{-1} .
- (2) The illuminance of the lamp is 756 lx .
- (3) Compared to other 12 W compact fluorescent lamps, the efficacy of this lamp is above average.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

- (2023) 3.2 In the room shown below, a lamp fixed at the centre of the ceiling is the only light source. It is 1.6 m from each wall.



Point X on the wall is at 1.2 m below the ceiling while point Y on the opposite wall is at 2.1 m below the ceiling. Find the ratio $\frac{\text{illuminance on the wall around point } X}{\text{illuminance on the wall around point } Y}$. Neglect any light reflecting off the ceiling, the floor and the walls.

- A. 0.43
B. 0.76
C. 1.32
D. 2.30

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2023) 3.3 The information of heat transfer of a house is tabulated below:

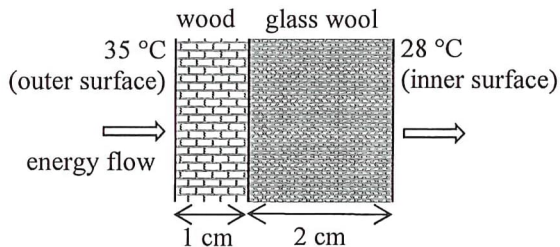
interface with the environment	walls	roof	windows
total area / m^2	102	70	x
annual average rate of heat gain per unit area / W m^{-2}	10.0	25.0	116

The Overall Thermal Transfer Value (OTTV) of the house is 22.4 W m^{-2} . Find the total area of windows, x .

- A. 7.8 m^2
- B. 9.3 m^2
- C. 11.6 m^2
- D. 23.9 m^2

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

- (2023) 3.4 A wall is composed of wood and glass wool of thickness 1 cm and 2 cm respectively. The outer surface of the wall is at a temperature of 35 °C while its inner surface is kept at 28 °C.



Given: thermal conductivity of wood = $0.15 \text{ W m}^{-1} \text{ K}^{-1}$

thermal conductivity of glass wool = $0.03 \text{ W m}^{-1} \text{ K}^{-1}$

Estimate the temperature at the wood and glass wool interface.

- A. 28.6 °C
- B. 30.0 °C
- C. 33.0 °C
- D. 34.4 °C

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2023)

- 3.5 An electric vehicle of mass 1500 kg is equipped with a regenerative braking system. When the brakes are applied, the speed of the vehicle is reduced from 70 km h^{-1} to 35 km h^{-1} . In this process 57.6 kJ of chemical energy is stored in its battery. Estimate the overall efficiency of the regenerative braking system.

A. 22%

B. 25%

C. 27%

D. 30%

A

B

C

D



(2023) 3.6 Solar panels usually have a dark appearance. This is because

- A. the cost to manufacture dark colour solar panels is lower.
- B. a transparent anti-reflection film is coated onto the top surface of the panel.
- C. the panels are painted dark so as to absorb more radiation.
- D. the panels are painted dark so that it won't be necessary to clean them as frequently.

A

B

C

D



(2023) 3.7 There are a total of 100 wind turbines used to generate electricity. Each turbine has blade length 6 m and an efficiency of 23%. Now they are replaced by N turbines of a newer model, with blade length 8 m and an efficiency of 28%. Estimate N assuming the total output power is almost unchanged.

A. 46

B. 62

C. 69

D. 91

A

B

C

D



(2023) 3.8 Which of the following statements about nuclear energy is/are correct ?

- (1) It is a kind of renewable energy.
- (2) If the magnitude of the total binding energy increases after a nuclear reaction, energy will be released.
- (3) Unstable nuclei tend to release energy through nuclear reactions to become more stable.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

A

B

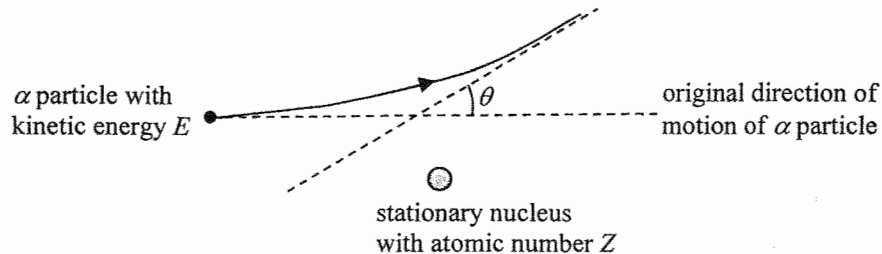
C

D



(2024) 2.1

Initially an α particle with kinetic energy E is far away from a stationary nucleus of atomic number Z . The α particle describes a curved trajectory making an angle of deviation θ with its original direction of motion as shown.



Which of the following change in E and Z would give the greatest θ ?

	E	Z	A	B	C	D
A.	increase	increase				
B.	increase	decrease	☐	☐	☐	☐
C.	decrease	increase				
D.	decrease	decrease				

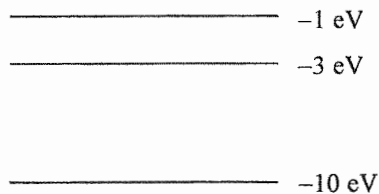
(2024) 2.2 When an electron collides with a gas atom and

- (1) if the collision is elastic, the electron transfers negligible kinetic energy to the atom.
- (2) if the atom is excited, all kinetic energy of the electron must be transferred to the atom.
- (3) if the atom is ionized, the atom must have absorbed the electron that collided with it.

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

- (2024)
2.3
- The figure shows the energy levels of an atom. Photons are emitted due to electron transitions between energy levels.



Which of the following statements is/are correct ? Given: the energy of photons in the visible spectrum lies between 1.78 eV and 3.10 eV

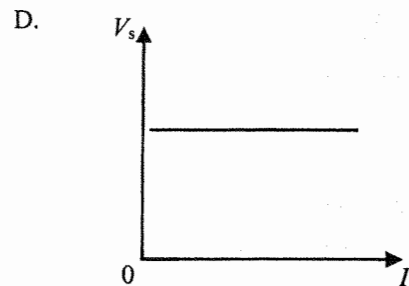
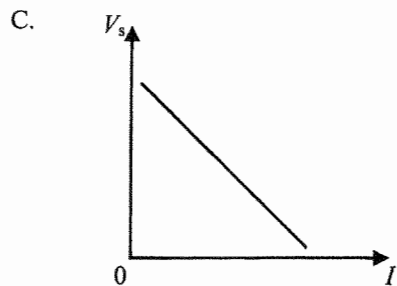
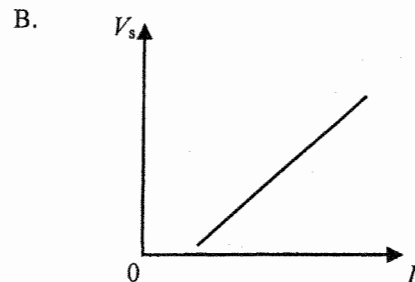
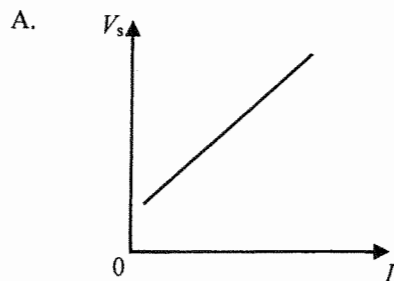
- (1) Some of these photons are in the visible spectrum.
- (2) At most, photons of two different frequencies can be emitted.
- (3) The photons with the longest wavelength come from transitions between -1 eV and -10 eV .

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2024)

2.4 In a photoelectric emission experiment, the variation of the stopping potential V_s of a metal with the intensity I of the monochromatic light used is studied. Which graph represents the experimental result ?



A B C D
☐ ☐ ☐ ☐

(2024) 2.5 Metal X and Y are illuminated in turns by monochromatic lights of wavelengths λ_1 and λ_2 . Both metals show photoemission for λ_1 while only metal Y shows photoemission for λ_2 . Which deduction below is correct ?

- A. $\lambda_1 > \lambda_2$ and the threshold frequency of X is higher.
B. $\lambda_1 > \lambda_2$ and the threshold frequency of Y is higher.
C. $\lambda_1 < \lambda_2$ and the threshold frequency of X is higher.
D. $\lambda_1 < \lambda_2$ and the threshold frequency of Y is higher.

A	B	C	D
☐	☐	☐	☐

- (2024) 2.6 The de Broglie wavelength of an electron of mass m is equal to the wavelength of a photon that has energy E . Which of the following expressions is/are equal to the momentum of the electron ?
(c = speed of light in vacuum)

(1) $\sqrt{2mE}$

(2) mc

(3) $\frac{E}{c}$

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

- (2024) 2.7 In a transmission electron microscope (TEM), emitted electrons accelerate in an electron gun and then pass through components (1) - (3) as shown in the block diagram below before being projected through a lens and forming an image on a screen.



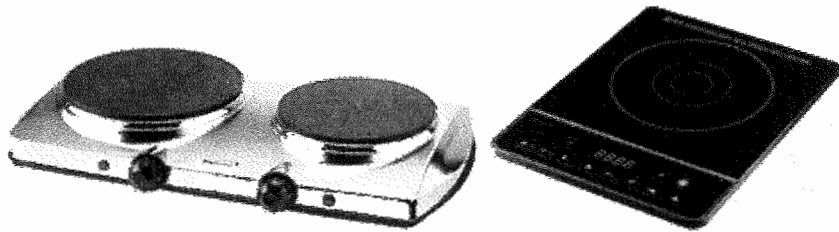
Which of the following combinations is correct ?

	(1)	(2)	(3)				
A.	condensing lens	objective lens	specimen	A	B	C	D
B.	condensing lens	specimen	objective lens	☐	☐	☐	☐
C.	objective lens	specimen	condensing lens				
D.	specimen	condensing lens	objective lens				

(2024) 2.8 Zinc oxide is a common ingredient in sunscreen products. Nanotechnology is often used in these products because

- | | A | B | C | D |
|---|-----------------------|-----------------------|-----------------------|-----------------------|
| A. zinc oxide in bulk is not able to block ultra-violet rays. | ☐ | ☐ | ☐ | ☐ |
| B. zinc oxide in bulk is harmful to the skin. | ☐ | ☐ | ☐ | ☐ |
| C. sunscreen products using nanosized zinc oxide particles are water repelling. | ☐ | ☐ | ☐ | ☐ |
| D. sunscreen products using nanosized zinc oxide particles look transparent when applied on the skin. | ☐ | ☐ | ☐ | ☐ |

- (2024) 3.1 After using an electric hotplate and an induction cooker to cook the same dish, it is found that the surface temperature of the electric hotplate is much higher.

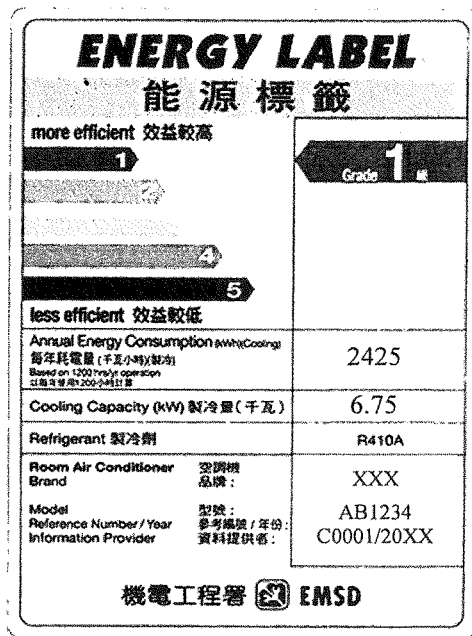


The most probable reason is that

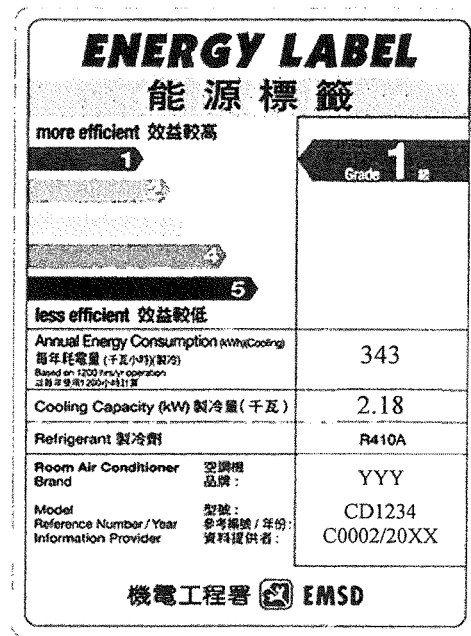
- A. the power of the electric hotplate is much higher.
- B. the electric hotplate heats up its plate and transfers thermal energy to the cooking utensil mainly by conduction.
- C. the process of electromagnetic induction does not produce thermal energy.
- D. the surface of the induction cooker is made of glass which is a poor conductor of heat.

A	B	C	D
☐	☐	☐	☐

(2024) 3.2 The energy labels of air-conditioners P and Q are shown below. The annual energy consumption (kW h) of each of them is based on 1200 hours of operation.



air-conditioner P



air-conditioner Q

Which one is (a) more energy efficient, (b) more suitable for a small room ?

more energy efficient more suitable for a small room

- | | | | | | | |
|----|-----|-----|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | Q | P | A | B | C | D |
| B. | P | P | ☐ | ☐ | ☐ | ☐ |
| C. | Q | Q | | | | |
| D. | P | Q | | | | |

(2024) 3.3 Which component(s), evaporator, compressor, condenser and expansion valve, is/are located in the indoor unit of a split-type air-conditioner ?

A. expansion valve, evaporator and condenser

B. expansion valve and compressor

C. evaporator

D. compressor

A

B

C

D



(2024) 3.4 Which of the factors below determine the thermal transmittance (U-value) of a wall ?

- (1) the thickness of the wall
- (2) the total area of the wall
- (3) the thermal conductivity of the wall's building material

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A	B	C	D
☐	☐	☐	☐

(2024) 3.5 Compared to common glass windows, which of the following about a double-glazed window is/are correct ?

- (1) Radiation penetrating the double-glazed window is significantly reduced.
- (2) The gas trapped in the double-glazed window has a much lower thermal transmittance than glass.
- (3) Double-glazed windows are not suitable for use in cold areas as more thermal energy would be lost through them.

- | | | | | | |
|----|------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | (2) only | A | B | C | D |
| B. | (3) only | ☐ | ☐ | ☐ | ☐ |
| C. | (1) and (2) only | | | | |
| D. | (1) and (3) only | | | | |

- (2024) 3.6 A container office of internal volume 37.5 m^3 is to be cooled from 33°C to 25°C by air-conditioners of total cooling capacity 8.80 kW . Estimate the time required if the container office has an exposed surface area of 57.5 m^2 and its Overall Thermal Transfer Value (OTTV) is 110 W m^{-2} .

Given: density of air $= 1.20 \text{ kg m}^{-3}$

specific heat capacity of air $= 1000 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

- A. 24 s
B. 41 s
C. 118 s
D. 145 s

A	B	C	D
☐	☐	☐	☐

(2024) 3.7 On a particular day, steady wind of speed 8.0 m s^{-1} blows normally to a wind turbine for 12 hours while for the rest of the day the wind speed is too low for operation. The maximum energy generated that day is equivalent to a day with a steady wind of speed of _____ throughout the day.

A. 6.3 m s^{-1}

B. 5.0 m s^{-1}

C. 4.0 m s^{-1}

D. 3.2 m s^{-1}

A

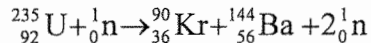
B

C

D



- (2024) 3.8 The equation below represents a typical fission reaction that occurs in a nuclear power plant.



Estimate the energy released when 1.00 kg of U-235 undergoes this fission reaction. Given that binding energies per nucleon of U-235, Kr-90 and Ba-144 are 7.59 MeV, 8.59 MeV and 8.26 MeV respectively. The mass of one mole of U-235 is 235 g.

- A. 2.38×10^{25} MeV
- B. 1.08×10^{26} MeV
- C. 4.56×10^{26} MeV
- D. 4.58×10^{26} MeV

- | A | B | C | D |
|-----------------------|-----------------------|-----------------------|-----------------------|
| ☐ | ☐ | ☐ | ☐ |

(2025) 2.1 In Thomson's plum pudding model, it is suggested that an atom is a positively charged sphere with electrons distributed evenly within it. Which of the following results of the α -scattering experiment implies that Thomson's model is **incorrect** ?

- (1) Most of the α particles pass straight through the thin gold foil.
- (2) Some α particles are deflected by large angles.
- (3) A few α particles are bounced back.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

A

B

C

D



(2025) 2.2 Which of the following about the Bohr atomic model is/are correct ?

- (1) Electrons are orbiting around the nucleus.
- (2) The attractive force between the nucleus and the electron provides the centripetal force for the electron's circular motion.
- (3) Photons are emitted when electrons jump from lower energy orbits to higher energy orbits.

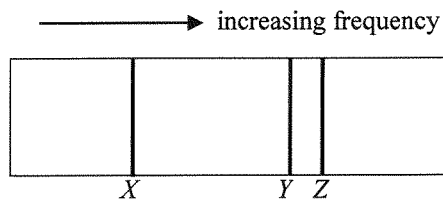
- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

A	B	C	D
☐	☐	☐	☐

(2025) 2.3 The energy levels of a hydrogen-like atom take the form $-\frac{36}{n^2}$ eV, where n is the quantum number. What is the energy level of its 2nd excited state ?

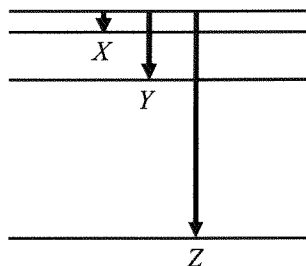
- A. -4 eV
- B. -9 eV
- C. -12 eV
- D. -18 eV

A	B	C	D
☐	☐	☐	☐

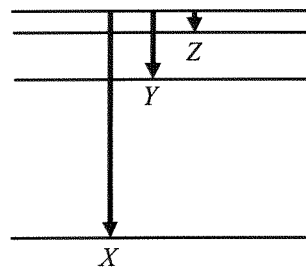


The above figure shows the spectrum obtained when the visible light from a discharge tube passes through a diffraction grating. Spectral lines X , Y and Z correspond to electron transitions between four energy levels. Which of the following best illustrates the electron transitions that give the spectral lines ?

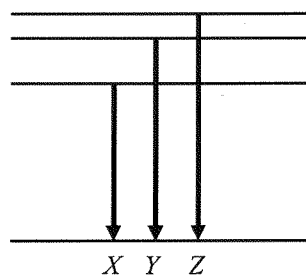
A.



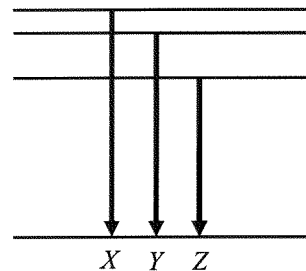
B.



C.



D.



A



B



C



D



(2025) 2.5 A beam of fast-moving electrons of velocity v behaves like a wave with wavelength λ and frequency f . Which of the following statements is/are correct ?

- (1) λ increases when v decreases.
- (2) The beam of electrons shows wave properties only with some discrete f .
- (3) The momentum associated with the electron beam is given by $\frac{\lambda}{\text{Planck constant}}$.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

- (2025) 2.6 Two point objects of separation 2.4×10^{-6} m are to be observed under a microscope using light of wavelength 550 nm. Which of the following combinations of aperture diameter and focal length of the microscope's objective lens is suitable ?

	aperture diameter	focal length	A	B	C	D
A.	5 mm	20 mm				
B.	6 mm	20 mm	☐	☐	☐	☐
C.	5 mm	24 mm				
D.	6 mm	24 mm				

(2025) 2.7 Which of the following statements about a scanning tunnelling microscope (STM) is/are correct ?

- (1) It uses a high energy electron beam instead of visible light.
- (2) STM gives a 3-dimensional image of a conductive surface.
- (3) The tunnelling current produced depends on the separation between the surface under investigation and the probe of the STM.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

A

B

C

D



(2025) 2.8 When a material is broken up into nanosized particles, which of the following statements is/are correct ?

- (1) Its chemical properties change but the physical properties remain unchanged.
- (2) Its shape changes but the atomic arrangement remains unchanged.
- (3) Its total surface area to volume ratio increases.

A. (1) only

B. (3) only

C. (2) and (3) only

D. (1), (2) and (3)

A

B

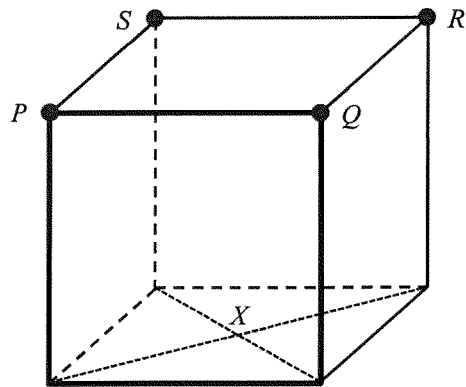
C

D



(2025) 3.1

In a room of dimensions $d \times d \times d$, identical point light sources, P , Q , R and S , are located at the four corners of the ceiling as shown. The luminous flux of each source is Φ .



Find the illuminance around point X on the floor in the middle of the room. Neglect reflections from the walls of the room.

A. $\frac{\Phi}{\pi d^2} \times \left(\frac{3}{2}\right)^{\frac{1}{2}}$

B. $\frac{\Phi}{\pi d^2} \times \left(\frac{3}{2}\right)^{\frac{3}{2}}$

C. $\frac{\Phi}{\pi d^2} \times \left(\frac{2}{3}\right)^{\frac{1}{2}}$

D. $\frac{\Phi}{\pi d^2} \times \left(\frac{2}{3}\right)^{\frac{3}{2}}$

A



B



C



D



(2025) 3.2 The end-use energy efficiency of microwave ovens is high because

- (1) microwaves only heat up the food without heating up the utensil directly.
- (2) microwaves do not heat up the nearby air much.
- (3) no energy is lost as heat in the circuit of the microwaves-generating component.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A

B

C

D



(2025) 3.3 Which of the following arrangements can reduce the Overall Thermal Transfer Value (OTTV) of a building envelope ?

- (1) using air-conditioners which are more energy-saving
- (2) using glass windows with low-emissivity coating
- (3) using building materials with greater U-value

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

A

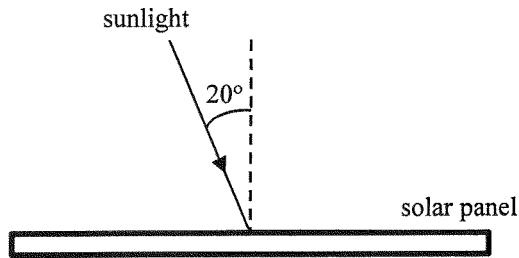
B

C

D



- (2025) 3.4 When incident sunlight makes an angle of 20° to the normal of a solar panel, the electric power generated is 100 W.

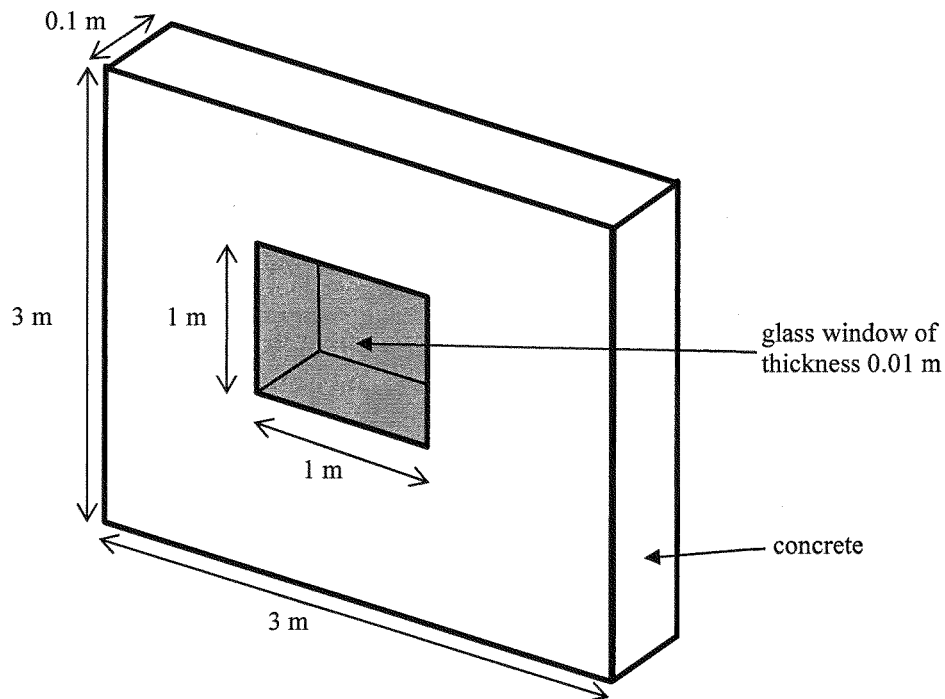


Assuming that the efficiency of the solar panel remains unchanged, find the electric power generated if the sunlight makes an angle of 30° to the normal of the panel and its intensity is reduced by 10%.

- A. $100 \times \frac{\sin 30^\circ}{\sin 20^\circ} \times 0.1 \text{ W}$
B. $100 \times \frac{\cos 30^\circ}{\cos 20^\circ} \times 0.1 \text{ W}$
C. $100 \times \frac{\sin 30^\circ}{\sin 20^\circ} \times 0.9 \text{ W}$
D. $100 \times \frac{\cos 30^\circ}{\cos 20^\circ} \times 0.9 \text{ W}$

- | A | B | C | D |
|-----------------------|-----------------------|-----------------------|-----------------------|
| ☐ | ☐ | ☐ | ☐ |

- (2025) 3.5 The figure shows a $3\text{ m} \times 3\text{ m}$ wall made of concrete of thickness 0.1 m and having a $1\text{ m} \times 1\text{ m}$ glass window of thickness 0.01 m .



The thermal conductivities of concrete and glass are $0.8\text{ W m}^{-1}\text{ K}^{-1}$ and $1.3\text{ W m}^{-1}\text{ K}^{-1}$ respectively. Find the overall thermal transmittance (U-value) of the wall.

- A. $20\text{ W m}^{-2}\text{ K}^{-1}$
B. $22\text{ W m}^{-2}\text{ K}^{-1}$
C. $73\text{ W m}^{-2}\text{ K}^{-1}$
D. $116\text{ W m}^{-2}\text{ K}^{-1}$

- A B C D
☐ ☐ ☐ ☐

- (2025) 3.6 Air-conditioners P and Q below operate respectively to cool two identical rooms under the same environmental conditions.

air-conditioner	input power / kW	coefficient of performance (COP)
P	1.6	2.2
Q	2.0	1.9

If P takes 400 s to cool down the room, estimate the time required for Q to cool down the other room to the same extent.

- A. 688 s
B. 432 s
C. 371 s
D. 276 s

A	B	C	D
☐	☐	☐	☐

- (2025) 3.7 Wind turbines P and Q below deliver roughly the same electric power output when they operate under the conditions tabulated.

wind turbine	blade length / m	wind speed / m s^{-1} (blowing normally to the turbine)
P	23	6.2
Q	15	8.0

The overall efficiency of electric power generation

- A. is roughly the same for both turbines.
B. is larger for P .
C. is larger for Q .
D. of the turbines cannot be compared as the maximum power available for tapping from the wind is not the same.

A	B	C	D
☐	☐	☐	☐

(2025) 3.8 What are the functions of pressurised water in the core of a pressurised water reactor ?

- (1) It acts as a moderator.
- (2) It acts as a coolant.
- (3) It turns into steam and drives the turbine.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

A

B

C

D



(2026) 2.1 Which of the following is/are the feature(s) of Rutherford's atomic model ?

- (1) All the charges are evenly distributed inside the atom.
- (2) Most of the mass is concentrated in a small nucleus at the centre of the atom.
- (3) The electrons revolve around the nucleus in certain discrete circular orbits.

A.	(2) only	A	B	C	D
B.	(3) only	☐	☐	☐	☐
C.	(1) and (2) only				
D.	(1) and (3) only				

(2026) 2.2 Monochromatic light X is incident on metal P and monochromatic light Y is incident on metal Q . Electrons are emitted from both metals through the photoelectric effect, with the same stopping potential of 1 V. However, when light X is incident on metal Q , no electrons are emitted. What occurs when light Y is incident on metal P ?

- A. No electrons are emitted.
- B. Electrons are emitted and the stopping potential is smaller than or equal to 1 V.
- C. Electrons are emitted and the stopping potential is larger than 1 V but smaller than 2 V.
- D. Electrons are emitted and the stopping potential is larger than or equal to 2 V.

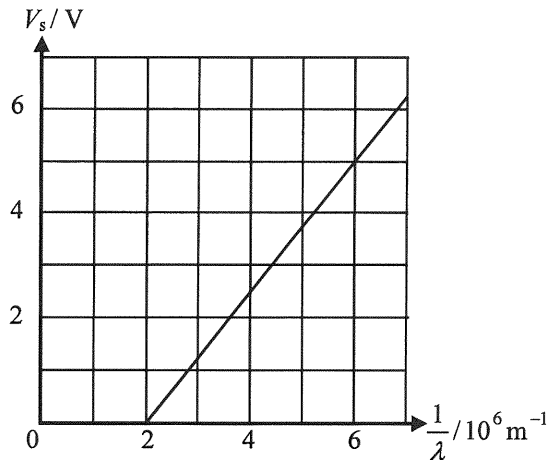
A	B	C	D
☐	☐	☐	☐

(2026) 2.3 Which of the following facts is/are required to account for the photoelectric effect ?

- (1) The energy of light is quantised.
- (2) The energy level of an atom is quantised.
- (3) Moving electrons show wave properties.

A.	(1) only	A	B	C	D
B.	(3) only	☐	☐	☐	☐
C.	(1) and (2) only				
D.	(2) and (3) only				

- (2026) 2.4 In a photoelectric experiment, lights of different wavelengths λ are shone on a metal surface in turns. The graph of the stopping potential V_s against $\frac{1}{\lambda}$ is shown below.



The work function of the metal is

- A. $2.0 \times 10^6 \text{ J}$.
- B. $3.2 \times 10^{-13} \text{ J}$.
- C. $4.0 \times 10^{-19} \text{ J}$.
- D. $1.3 \times 10^{-27} \text{ J}$.

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2026) 2.5 A hydrogen atom in ground state can absorb a photon of energy E and be excited to the first excited state. If another photon of energy $2E$ is incident on the atom in ground state, which of the following statements is correct ?

- A. The atom can absorb the photon and be excited to the second excited state.
- B. The atom can absorb the photon and be excited to an energy level higher than the second excited state.
- C. The atom can absorb the photon and become ionised.
- D. The atom cannot absorb the photon at all.

A

B

C

D



(2026) 2.6 A specimen slide is observed using an optical microscope under red light. The resolving power can be increased by

- (1) increasing the brightness of the light.
- (2) using an objective lens with a larger aperture.
- (3) replacing the red light by blue light.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

(2026) 2.7 In a transmission electron microscope (TEM), an electron beam is produced by an electron gun. The de Broglie wavelength of the electron beam is λ when the accelerating voltage is V . If the accelerating voltage is increased to $2V$, what is the de Broglie wavelength of the electron beam ?

- | | | | | | |
|----|-----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | $\frac{1}{\sqrt{2}}\lambda$ | A | B | C | D |
| B. | $\frac{1}{2}\lambda$ | ☐ | ☐ | ☐ | ☐ |
| C. | $\sqrt{2}\lambda$ | | | | |
| D. | 2λ | | | | |

(2026) 2.8 Carbon nanotubes

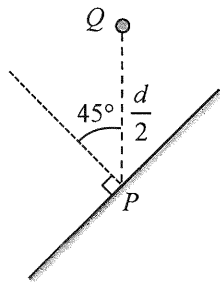
- (1) are stronger but less dense than steel.
- (2) are good thermal insulators.
- (3) can be good electrical conductors.

A.	(2) only	A	B	C	D
B.	(3) only	☐	☐	☐	☐
C.	(1) and (2) only				
D.	(1) and (3) only				

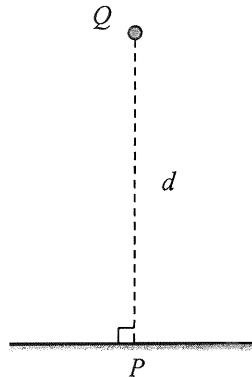
(2026)

3.1

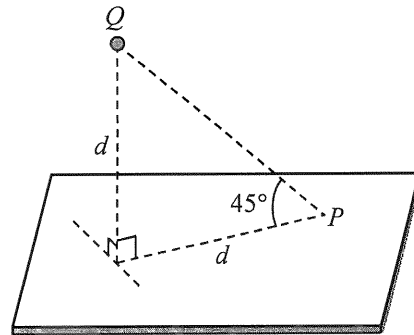
A point light source Q is placed above a plane surface in the following three different situations. Arrange the illuminance around point P in ascending order.



(1)



(2)



(3)

- A. 2, 3, 1
 B. 1, 2, 3
 C. 3, 1, 2
 D. 3, 2, 1

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| A | B | C | D |
| ☐ | ☐ | ☐ | ☐ |

(2026) 3.2 Which of the following descriptions about microwave ovens and induction cookers is/are correct ?

- (1) An induction cooker generates a rapidly changing magnetic field, which induces eddy currents to form at the base of compatible cookware, and thus heating the cookware.
- (2) The end-use energy efficiency of a microwave oven is higher than that of an induction cooker.
- (3) In a microwave oven, the glass door contains a vacuum layer that prevents microwaves from passing through and exposing the users.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

A

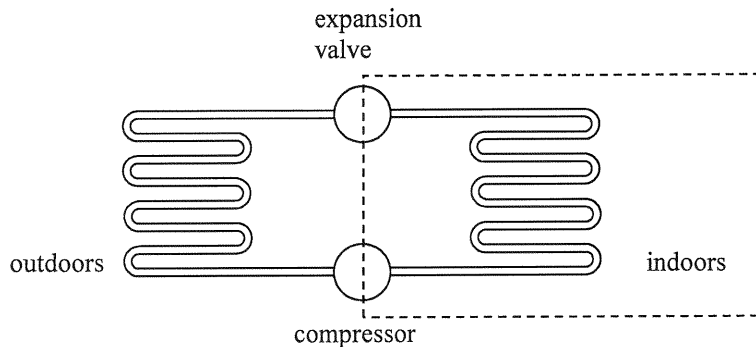
B

C

D



(2026) 3.3 A typical window-type air-conditioner is shown below.



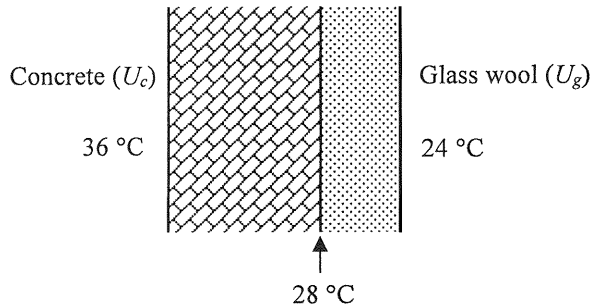
Which of the following statements is/are correct ?

- (1) The amount of heat released to outdoors is equal to the amount of heat removed from indoors.
- (2) The condensing coil is placed outdoors.
- (3) The liquid refrigerant in the evaporator coil absorbs heat from air and evaporates.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

A	B	C	D
☐	☐	☐	☐

- (2026) 3.4 A wall is composed of a concrete layer and a glass wool layer of different thicknesses. The two sides of the wall are respectively maintained at 36°C and 24°C as shown. The temperature at the concrete and glass wool interface is 28°C . Find the ratio of the U-value of the concrete layer (U_c) to that of the glass wool layer (U_g).



- A. 1 : 1
B. 1 : 2
C. 2 : 1
D. Cannot be determined as the thicknesses of the layers are not given.

A	B	C	D
☐	☐	☐	☐

(2026) 3.5 A wind turbine delivers a power of 1.5 kW at a wind speed of 5.0 m s^{-1} blowing normal to its blades. Take the efficiency of the turbine as 35% and the density of air as 1.23 kg m^{-3} . Estimate the length of the blade.

- A. 2.5 m
- B. 3.1 m
- C. 4.2 m
- D. 7.5 m

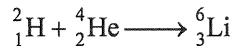
A	B	C	D
☐	☐	☐	☐

(2026) 3.6 A solar panel of area 5.0 m^2 , which has an efficiency of 12%, is installed on the roof of a building. The output power of the panel is 0.5 kW when sunlight makes an angle of 20° to the normal of the panel. Estimate the percentage of solar power absorbed by the atmosphere. Given: solar constant = 1366 W m^{-2}

- A. 20%
- B. 23%
- C. 35%
- D. 92%

A	B	C	D
☐	☐	☐	☐

(2026) 3.7 The equation below represents a nuclear reaction.



Given: Binding energy per nucleon of ${}^2_1\text{H} = 1.1 \text{ MeV}$

Binding energy per nucleon of ${}^4_2\text{He} = 7.1 \text{ MeV}$

Binding energy per nucleon of ${}^6_3\text{Li} = 5.3 \text{ MeV}$

Which of the following statements about this nuclear reaction is correct ?

A. 1.2 MeV energy is absorbed.

B. 1.2 MeV energy is released.

C. 2.9 MeV energy is absorbed.

D. 2.9 MeV energy is released.

A

B

C

D



(2026) 3.8 Which of the following is/are the impact(s) on the environment of a nuclear power plant ?

- (1) Warm water discharged into rivers and seas causes thermal pollution.
- (2) The release of sulphur dioxide causes acid rain.
- (3) The release of greenhouse gases during the nuclear reaction causes global warming.

- | | | | | | |
|----|------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| A. | (1) only | A | B | C | D |
| B. | (3) only | ☐ | ☐ | ☐ | ☐ |
| C. | (1) and (2) only | | | | |
| D. | (2) and (3) only | | | | |